



Machine Learning Using SAS® Viya®

Slides and Notes

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Machine Learning Using SAS® Viya® Slides and Notes

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Course Overview



Welcome from the Instructors

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Practicing in this Course (Required)

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Lesson 1: Getting Started with Machine Learning and SAS Viya



1.0 Lesson Overview



Lesson Overview

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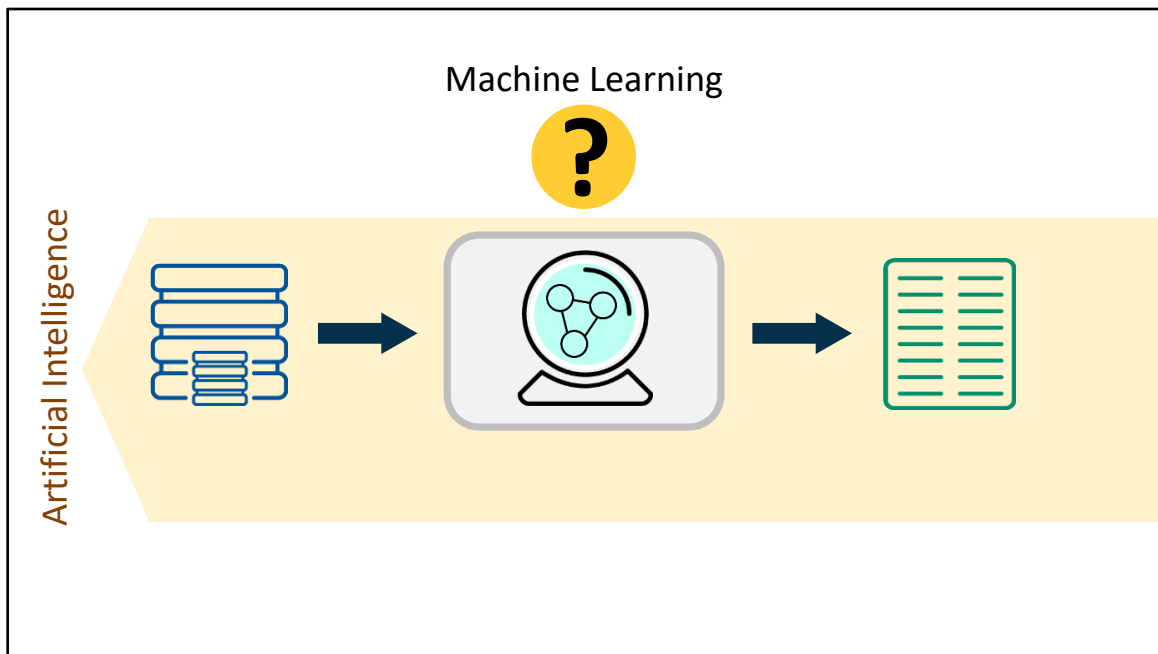
1.1 Machine Learning in Business Decision Making



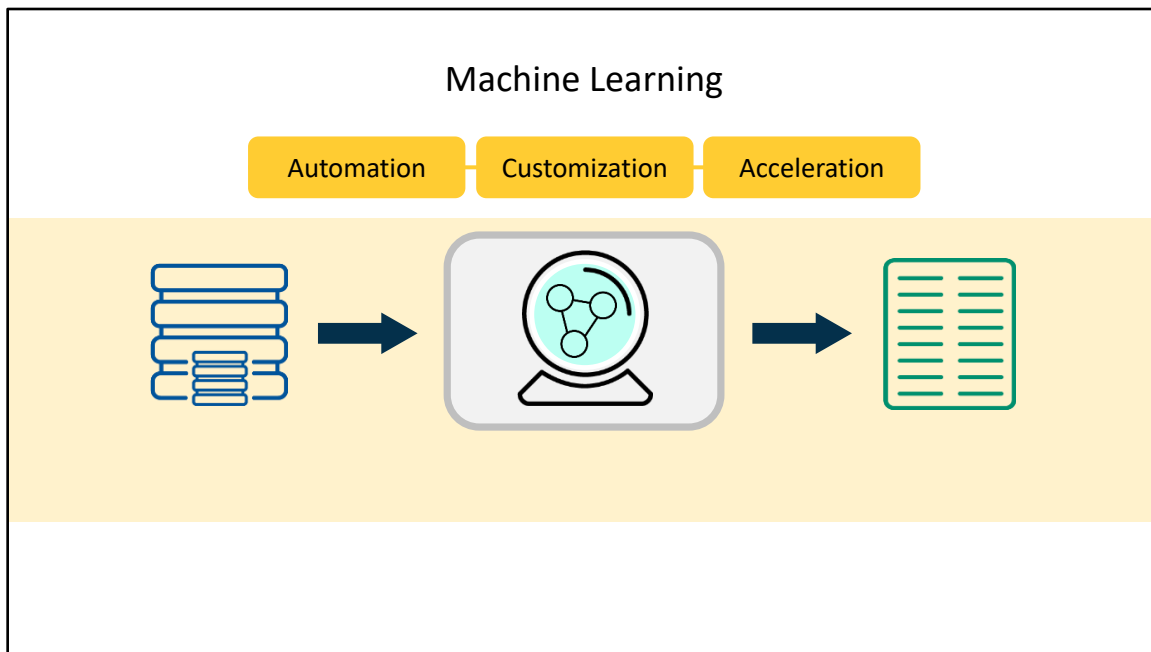
Machine Learning in SAS Viya



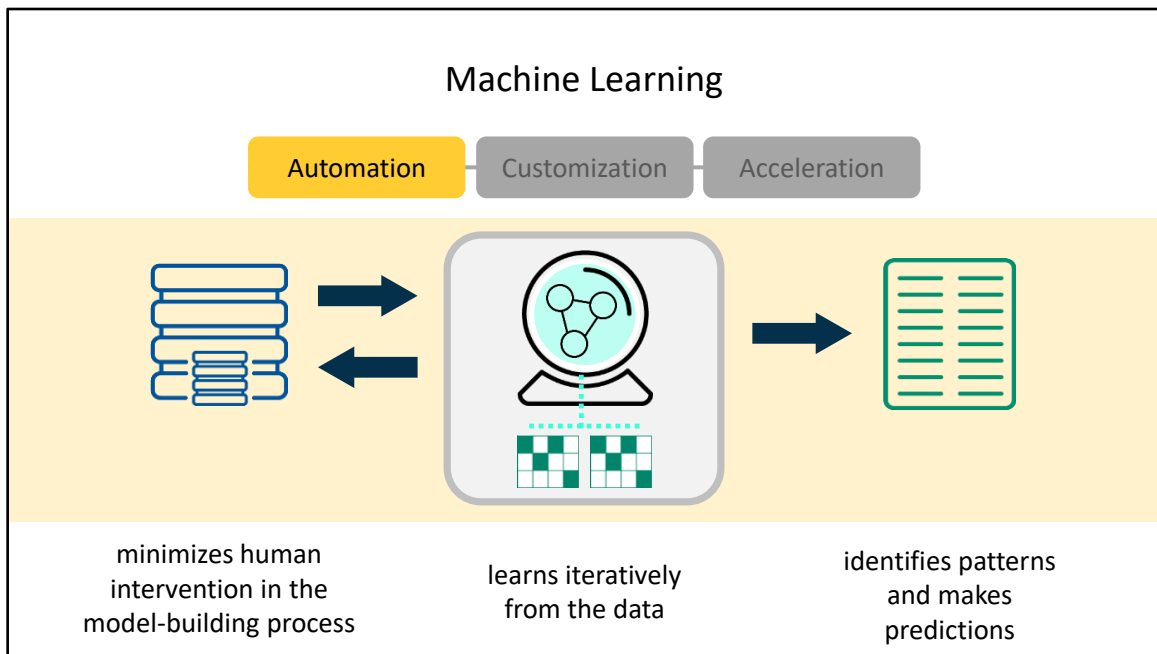
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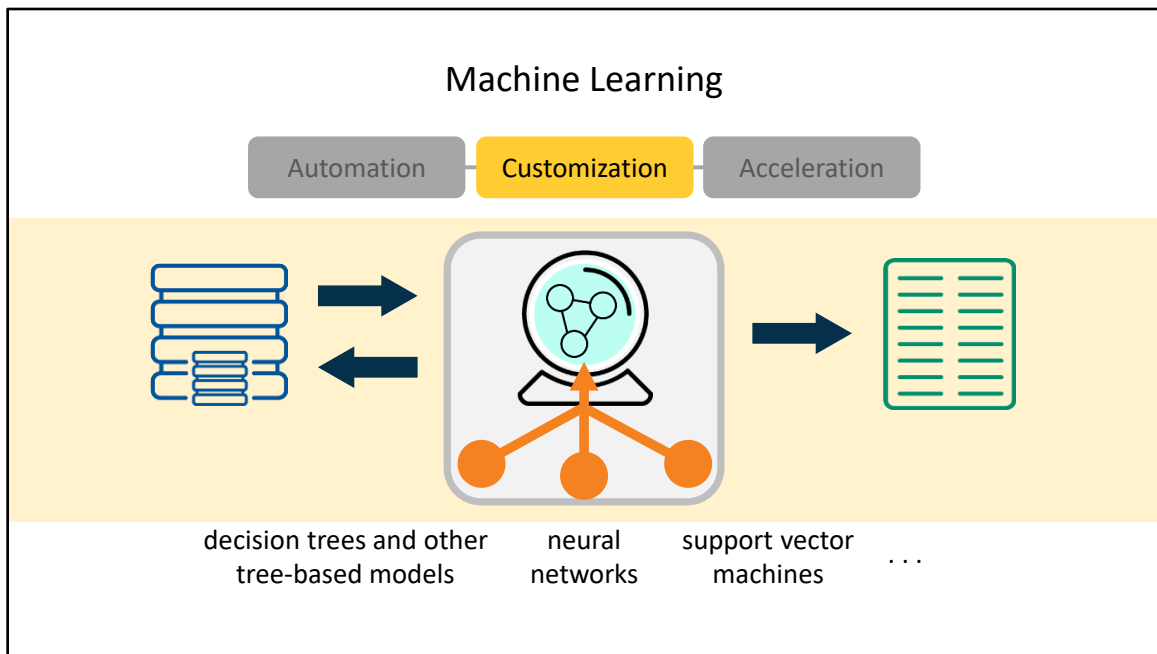
So, what is machine learning? Machine learning is a branch of artificial intelligence that builds systems that learn iteratively from data and make predictions.



Three main characteristics of machine learning are automation, customization, and acceleration.



Machine learning is largely an **automated** process. Model building requires minimal human intervention. Click Run and the system learns iteratively from the data, based on algorithms instead of programming. With each pass through the data, the system identifies patterns and makes predictions, and continues to learn and improve. A neural network is an example of a machine learning model. In a neural network, the weights start close to zero. With each pass through the data, the neural network learns more and refines the weights.



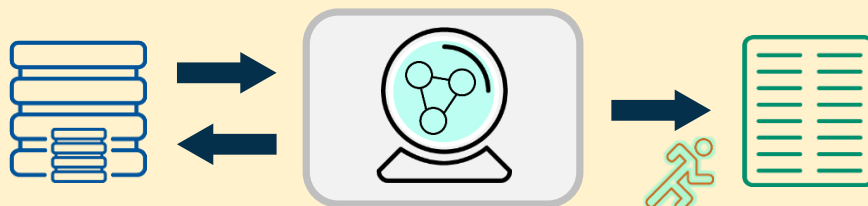
The machine learning process is highly **customized**. Machine learning can draw from many algorithms for training the data, depending on the requirements of the situation. In this course, you learn about several types of machine learning models: decision trees and other tree-based models, neural networks, and support vector machines.

Machine Learning

Automation

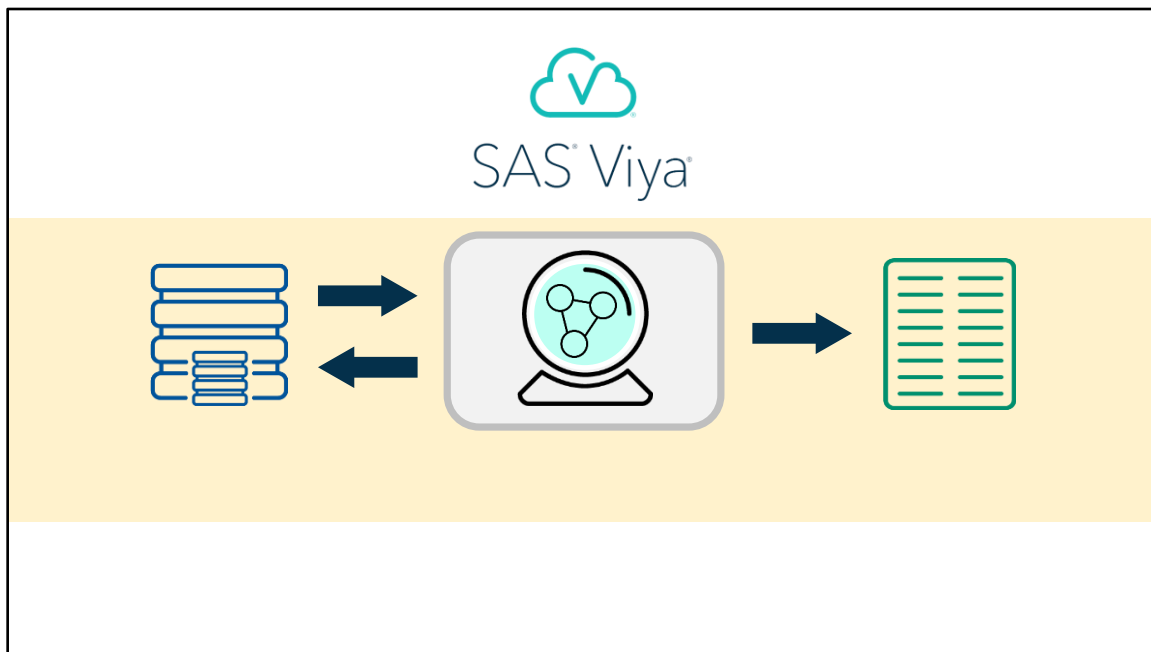
Customization

Acceleration

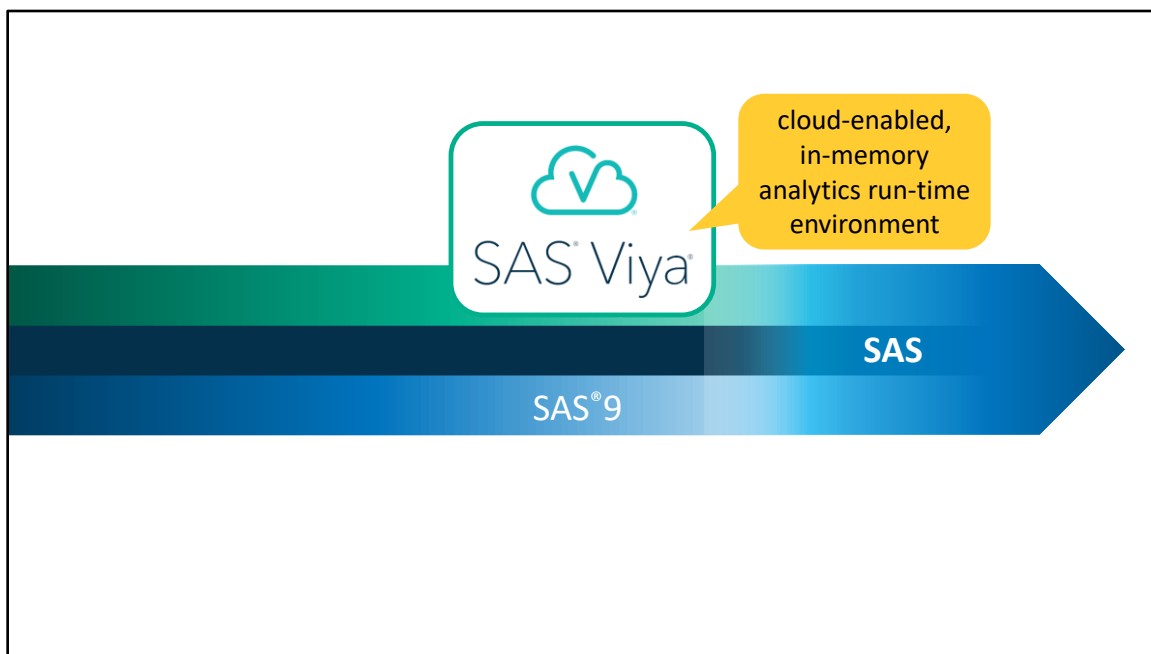


fast response time for sophisticated analytics
applied to data of any size or complexity

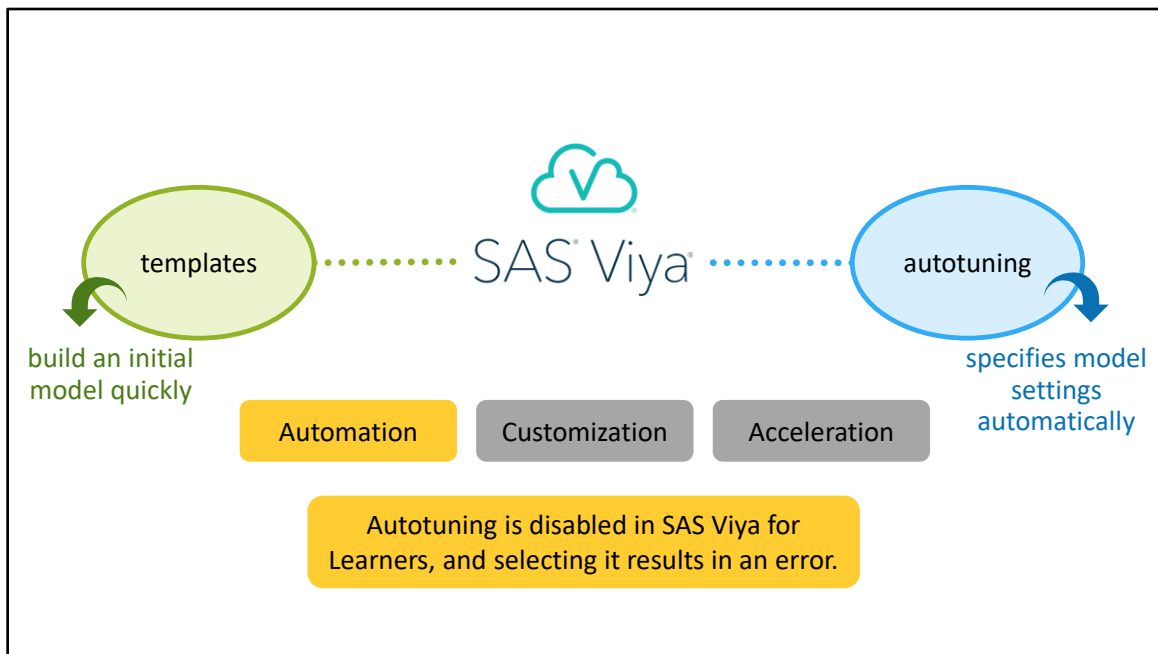
Finally, machine learning **accelerates** the process of running sophisticated analytics on big data so that results are generated quickly.



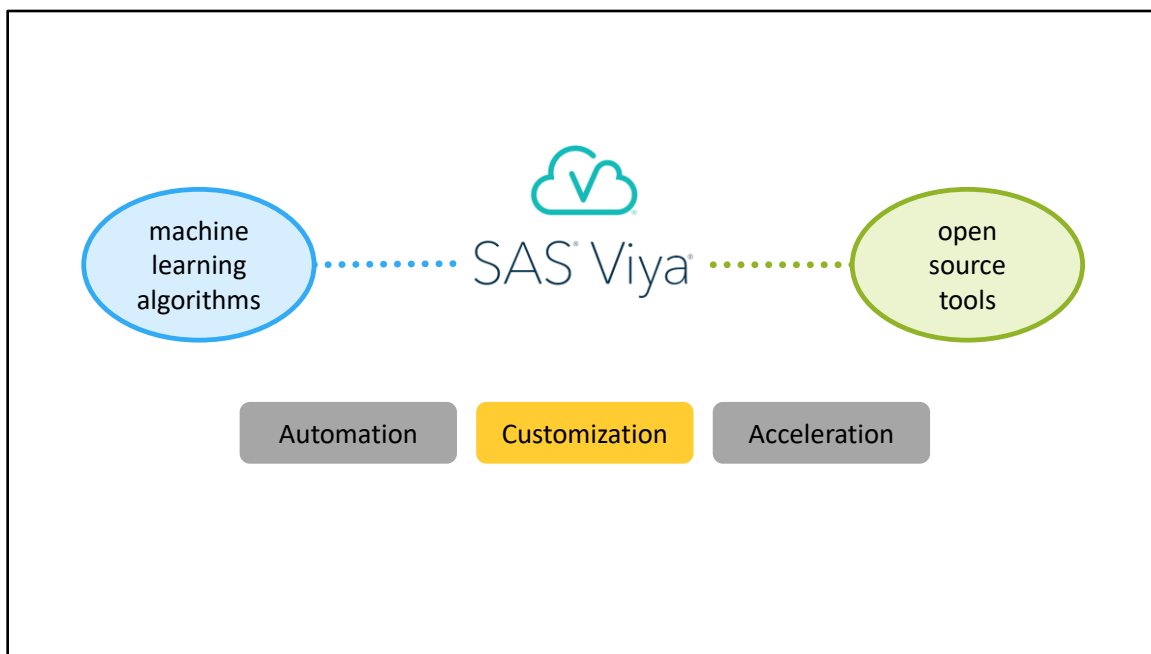
Machine learning shares many approaches with other related fields. SAS Viya offers the tools that you need for machine learning.



SAS Viya is a cloud-enabled, in-memory analytics run-time environment that seamlessly scales for data of any size, type, speed, and complexity.



Templates enable you to build an initial model quickly. The autotuning feature automates the process of specifying model settings. You can then apply your business knowledge and experience to improve on the automated results.



And you can **customize** your machine learning models by combining the state-of-the-art machine learning algorithms in SAS Viya with open source tools.

Analytics Life Cycle



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Analytics Life Cycle

Analysis Goal

Business
Challenge

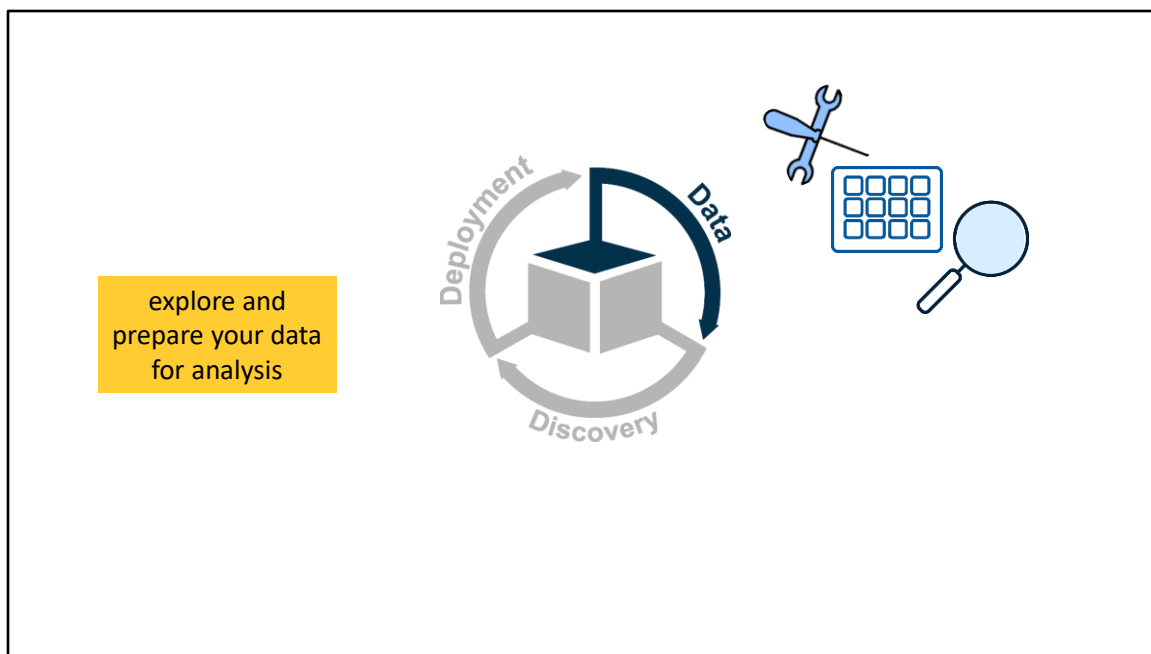
The analytics process always starts with a business challenge. In response, the organization identifies a precise and measurable analysis goal.



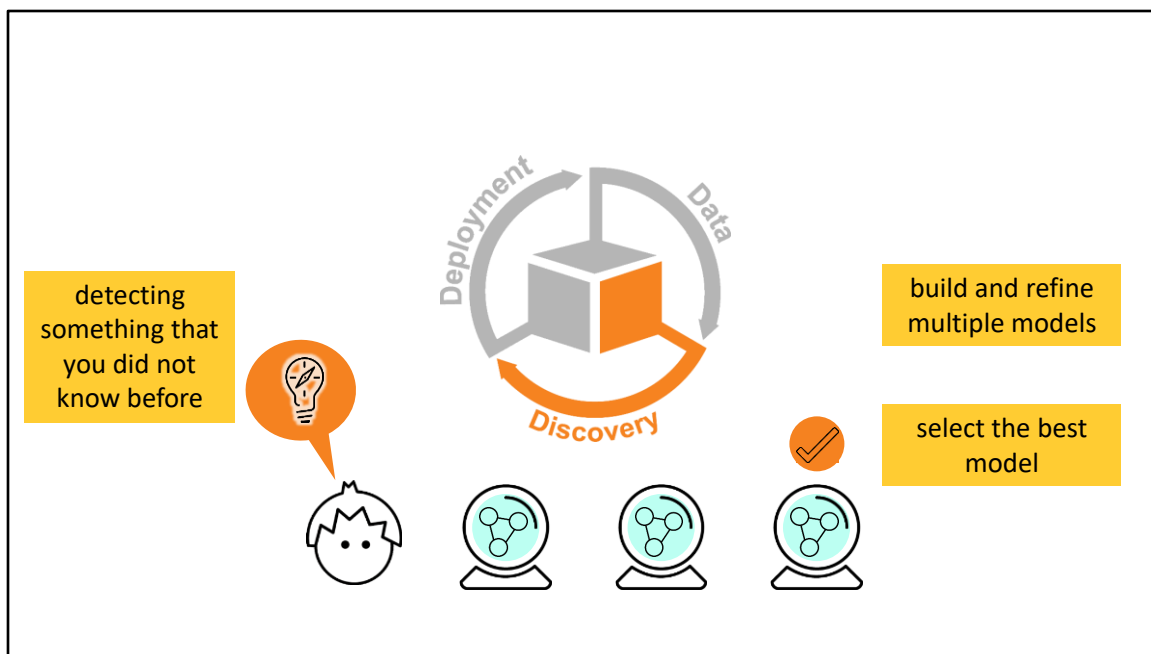
The analytics life cycle represents a series of activities to extract value from raw data.

The analytics life cycle represents a series of activities whose goal is to extract value from raw data. Of course, the definition of “value” is specific to each organization’s goals and objectives.

The analytics life cycle has three phases: Data, Discovery, and Deployment.



Data are the foundation of everything you do. At the Data phase, you explore and prepare your data for analysis.



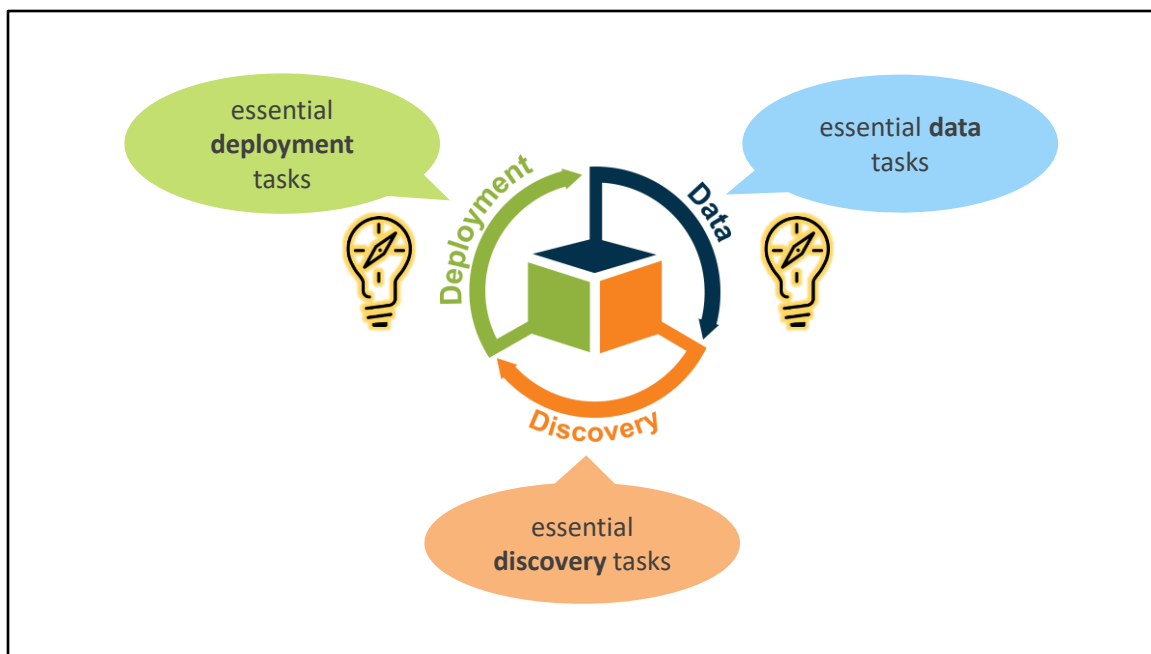
Discovery is the act of detecting something that you did not know before. You build and refine multiple models with the goal of selecting the best model for your analysis.



Scoring



Deployment is where you put the model to work. You apply the model to new data, which is a process called scoring.



In fact, the value of machine learning is evident throughout the analytics life cycle, with actionable insights at each phase.

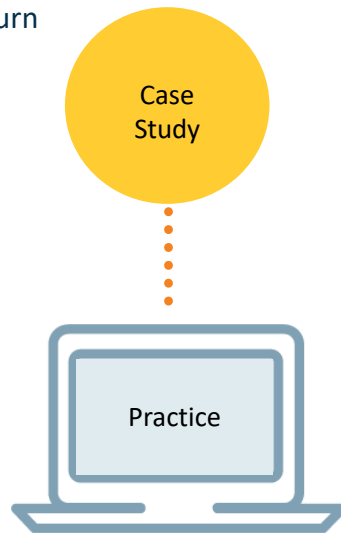
As you create machine learning models with SAS Viya tools in this course, you'll learn to perform the essential tasks that are defined for each phase. It's good to know that the analytics life cycle is flexible. When you apply what you've learned to your own projects in the future, you might do things in a different sequence.

Case Study: Customer Churn

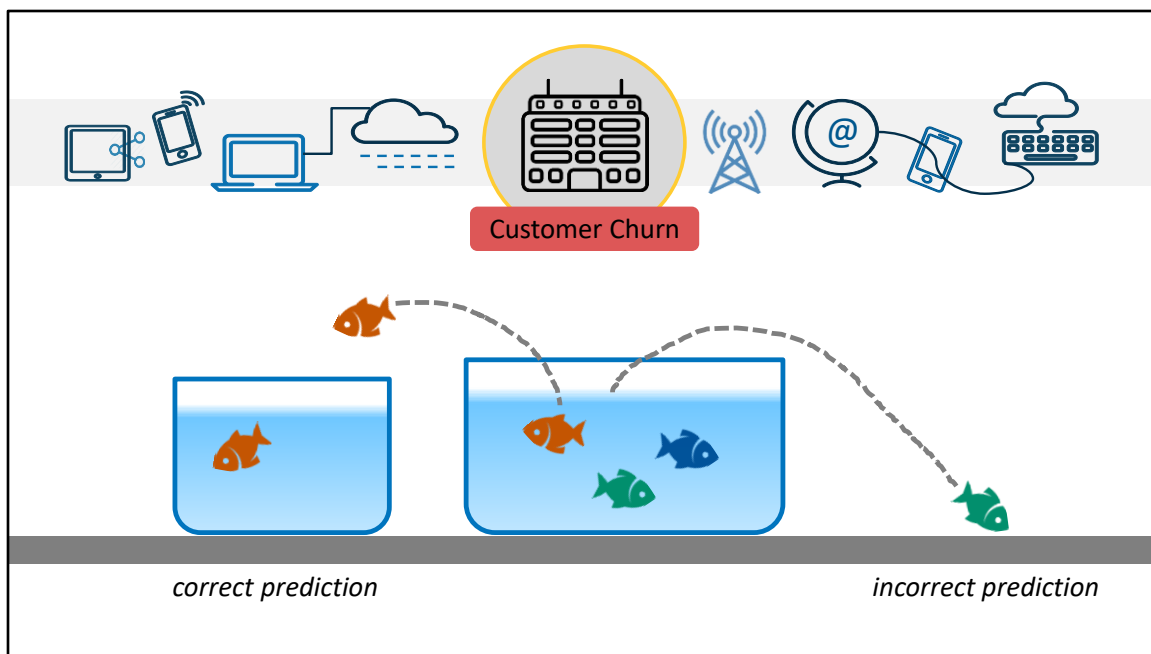


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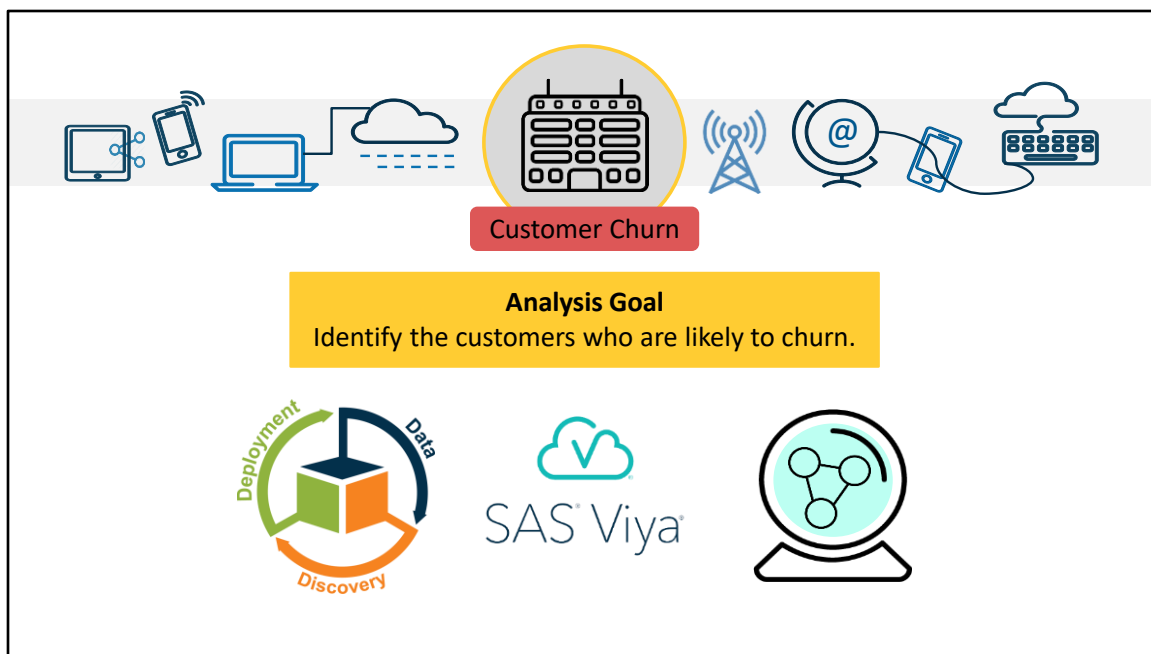
Case Study: Customer Churn



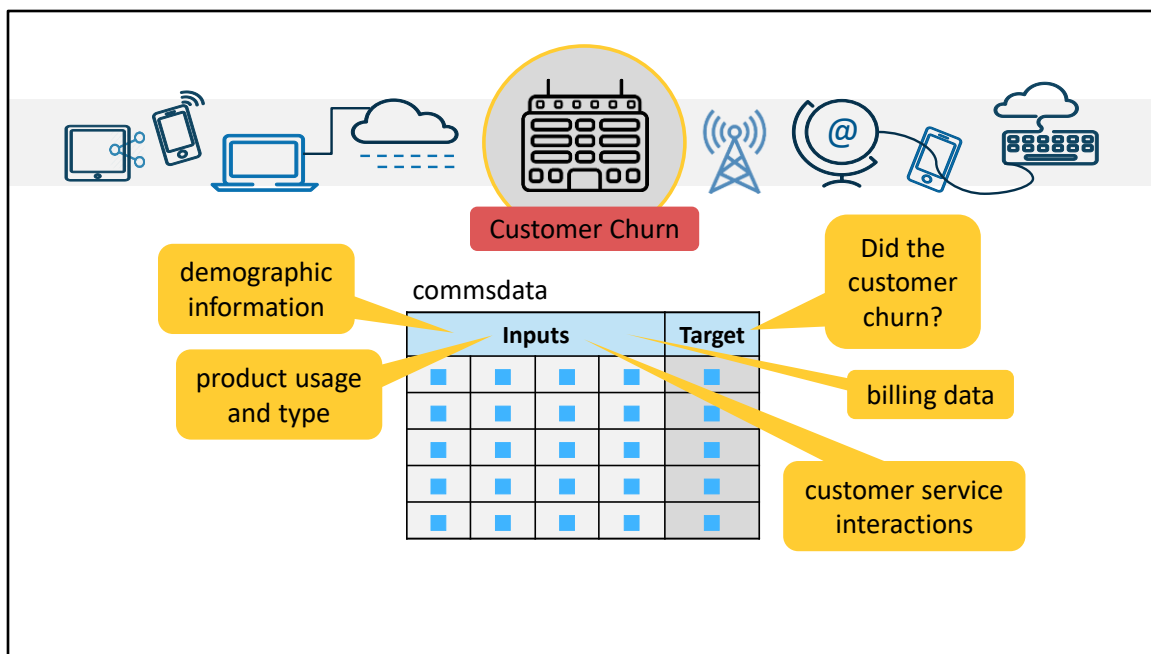
Throughout this course, you will practice what you learn with a case study.



A fictitious telecommunications company has a problem with customer churn. When existing customers or subscribers stop doing business or otherwise end a relationship with a company, they are said to churn, and this is bad for the bottom line. It is more costly to attract new customers than to keep existing customers.



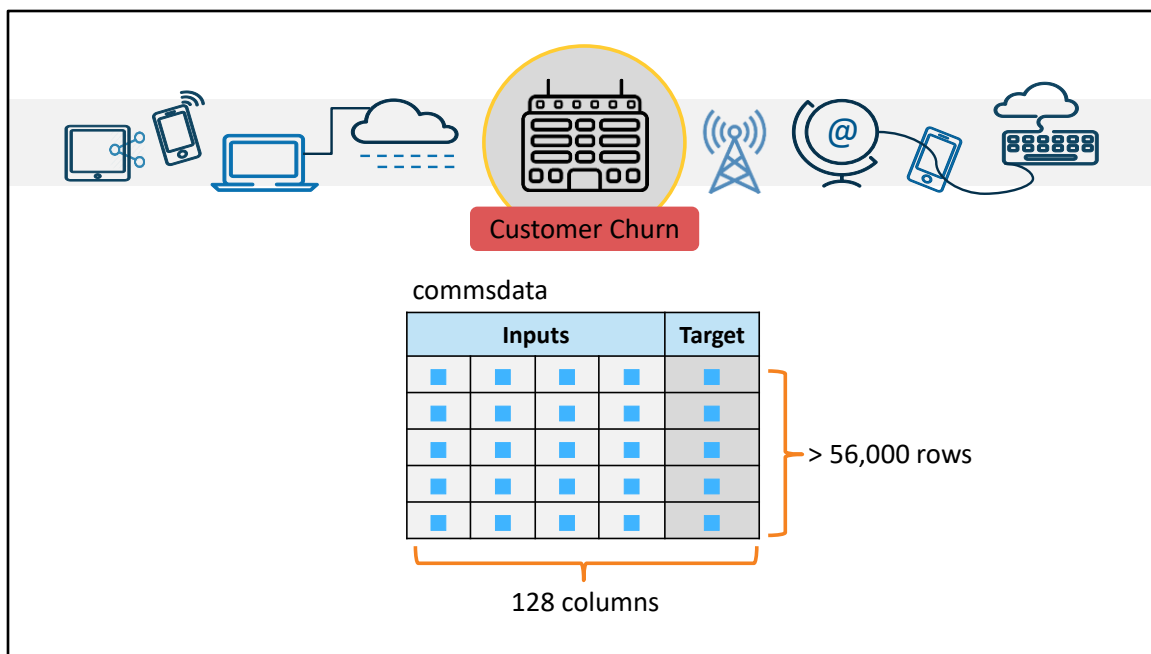
Many companies use analytics to identify customers at risk for churn and to understand how to retain them. Telecommunications providers routinely use predictive models to reduce the churn rate for post-paid subscribers, or customers who have a contract. This is the analysis goal for our case study. The results of your analysis could help management deploy effective retention and loyalty programs.



A little more about your case study: The company has raw data in a SAS table named `commsdata`, that contains information about customer behavior.

The input variables include demographic information, product usage and type, billing data, and interactions with customer service.

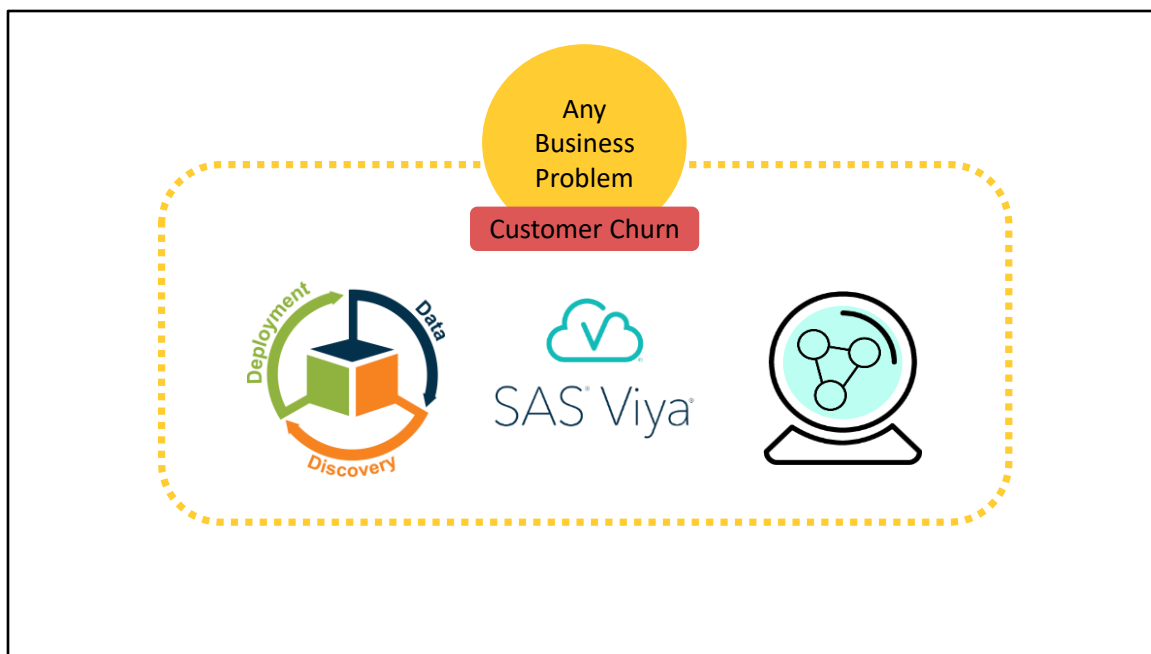
There is also a binary variable that indicates whether customers churned, and this is the target that you are trying to predict.



The commsdata data set contains 128 columns and more than 56,000 rows.

Don't worry if some of the terminology is unfamiliar right now—we'll explain everything as you go through the course.

Your assignment is to use this data set to create supervised models that predict whether customers will churn. Your instructors will use the same data to teach concepts—and your practice is to perform the same analyses as your instructor. Then, you will have some extra practice beyond what the instructor shows you in the demonstration videos.



Although this course uses customer churn to practice with machine learning models, you can apply what you learn here to many other business problems across a vast array of domains.

Applications of Prediction-Based Decision Making

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Case Study: Data Dictionary

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SAS Viya Applications Menu

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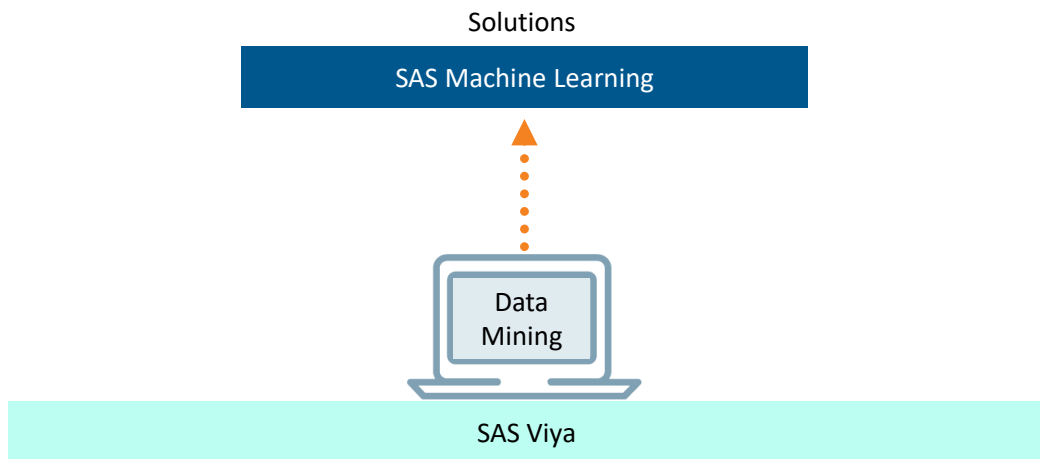


SAS Viya Tools for Machine Learning



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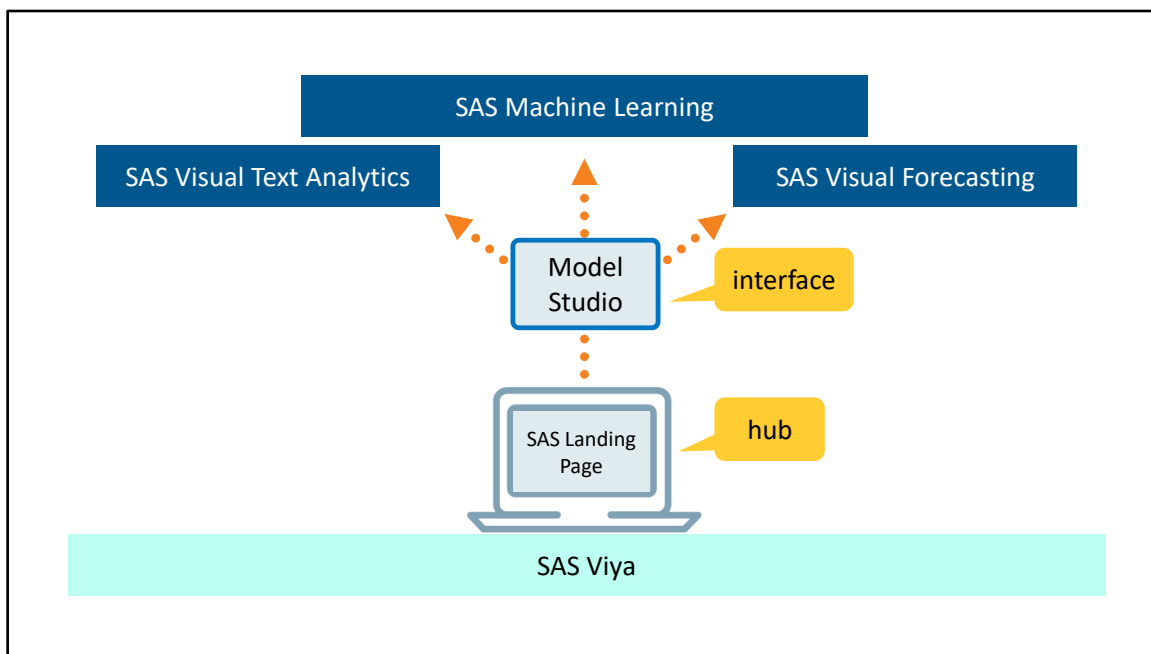
SAS Viya Tools for Machine Learning



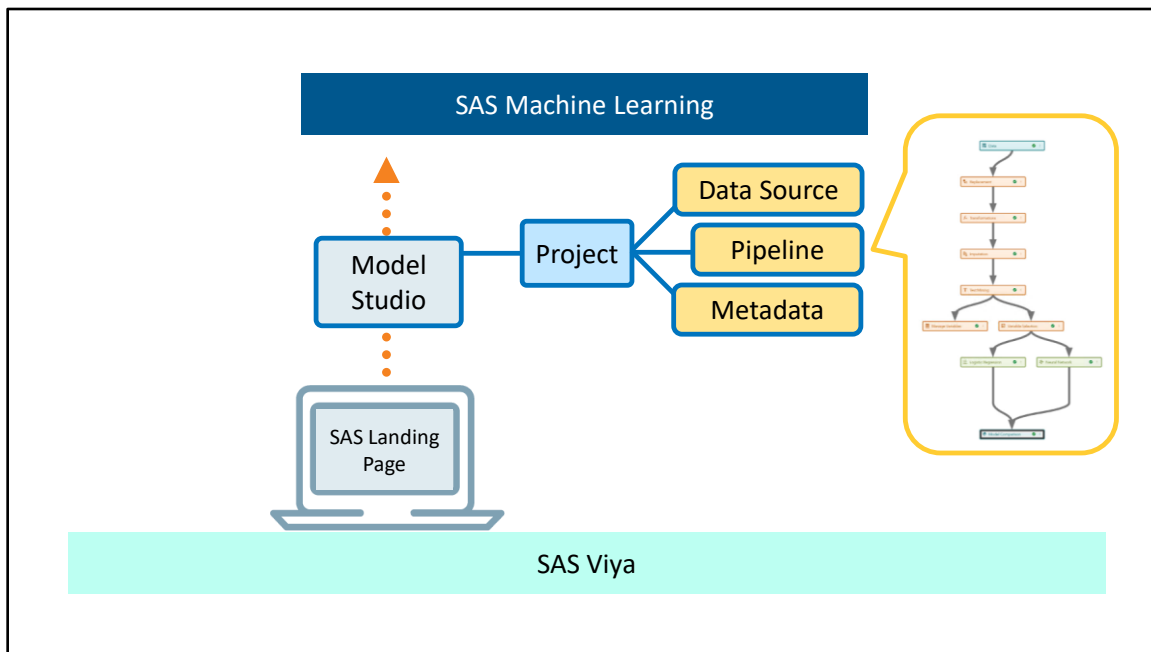
Here is a quick introduction to the SAS Viya tools that you learn about in this course, and how those tools work together.

The SAS Viya foundation supports a variety of software solutions for data mining.

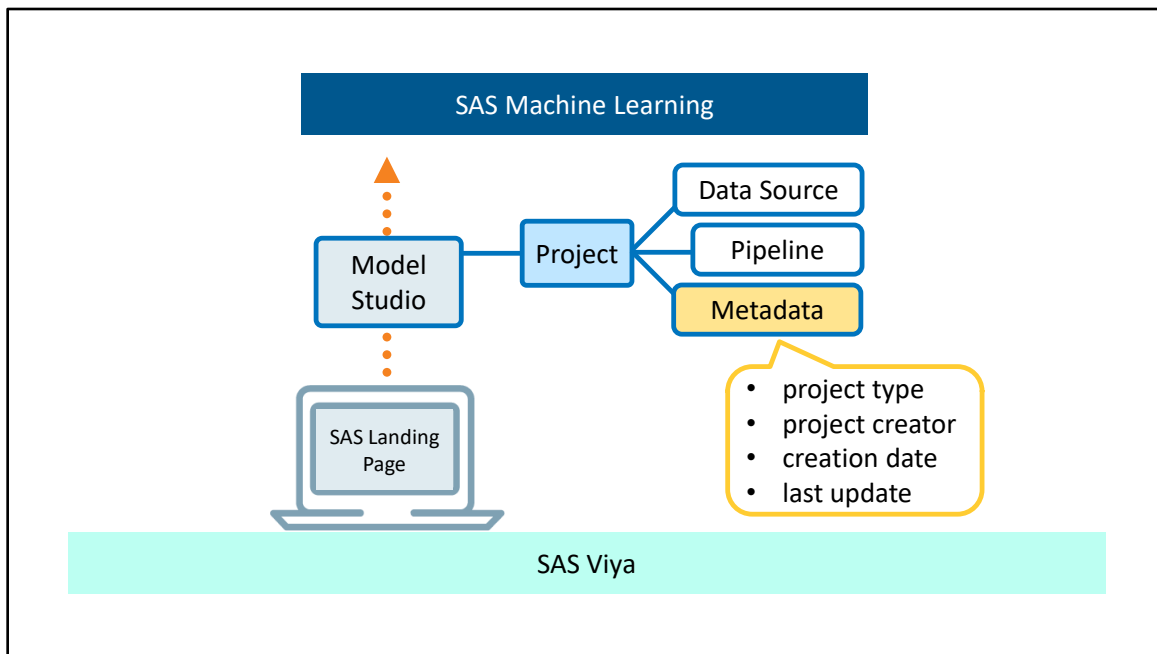
This course shows you how to use analytic data mining tools that are available in the SAS Machine Learning solution. These tools take advantage of SAS Viya programming and cloud processing environments for building predictive models and putting them into production.



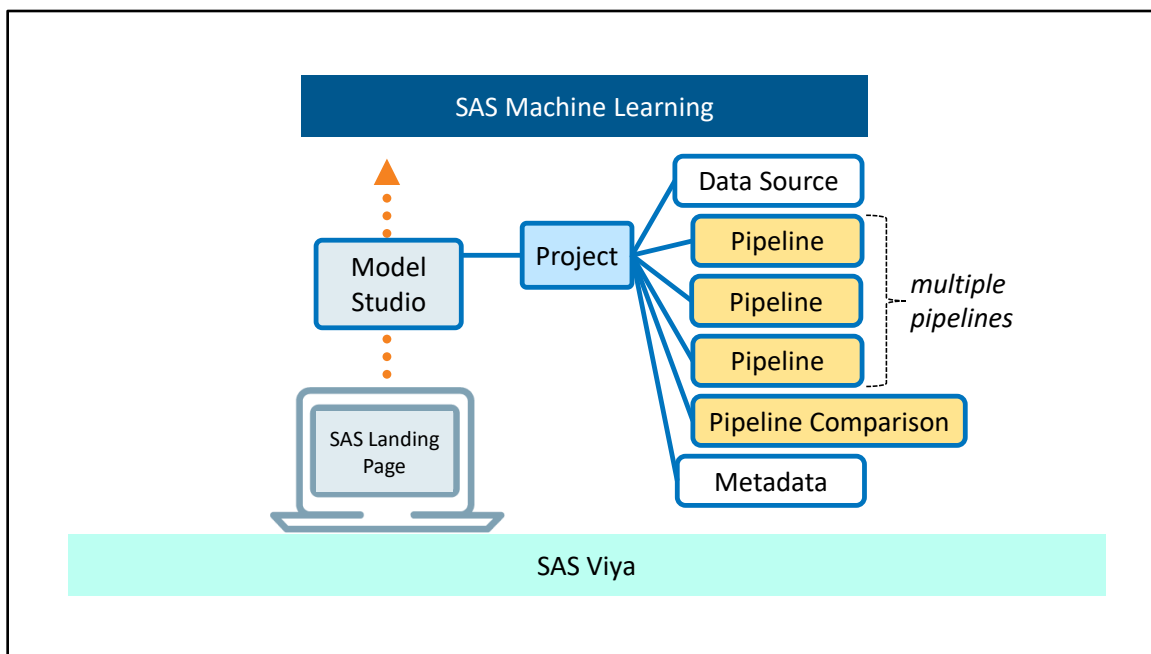
SAS Landing Page is the web-based hub (that is, the single point of access) for SAS Viya solutions. From SAS Landing Page, you access the features of SAS Machine Learning through Model Studio, a web-based visual interface. Model Studio is also an interface for two other SAS Viya solutions: SAS Visual Text Analytics and SAS Visual Forecasting. Your site's licensing agreement determines which solutions you can access from Model Studio.



In Model Studio, the main container for your analytic work is a project. A basic Model Studio project contains a data source, a pipeline that you create, and related project metadata. Remember that a pipeline is a structured workflow of analytic actions that is represented as a diagram. In the pipeline diagram, a node represents each action.



Some examples of project metadata are the project type, project creator, creation date, and last update.



A project can also have multiple pipelines and pipeline comparison results.

SAS Landing

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Demo: Creating a Project and Loading Data

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Question: Model Studio



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Question: Model Studio

Model Studio is a web-based platform that includes a suite of integrated data mining tools to facilitate end-to-end data mining analysis. Model Studio can be used with SAS Machine Learning in SAS Viya and two other visual tools: SAS Visual Text Analytics and SAS Visual Forecasting.

- a. True
- b. False

Answer: Model Studio

Model Studio is a web-based platform that includes a suite of integrated data mining tools to facilitate end-to-end data mining analysis. Model Studio can be used with SAS Machine Learning in SAS Viya and two other visual tools: SAS Visual Text Analytics and SAS Visual Forecasting.

- a. True
- b. False

CA: True

FB: Model Studio can be used with SAS Machine Learning in SAS Viya and two other visual tools: SAS Visual Text Analytics and SAS Visual Forecasting.

1.2 Supervised Prediction: Preparing the Data and Building the Initial Model



Predictive Modeling



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Predictive Modeling

Predictive Modeling

Supervised Prediction

Supervised Learning

Training Data

observations

Inputs				Target
■	■	■	■	■
■	■	■	■	■
■	■	■	■	■
■	■	■	■	■
■	■	■	■	■

training cases

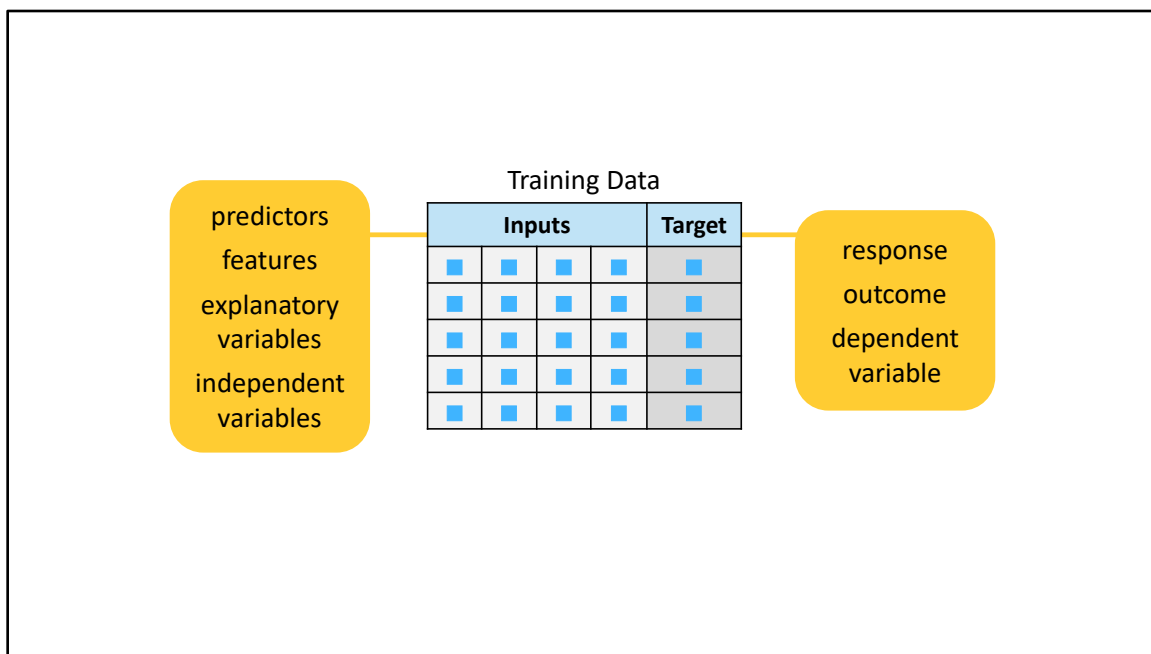
examples

instances

records

Predictive modeling (also known as *supervised prediction* or *supervised learning*) starts with a training data set.

The observations in a training data set are known as *training cases* (and are also known as *examples*, *instances*, or *records*).

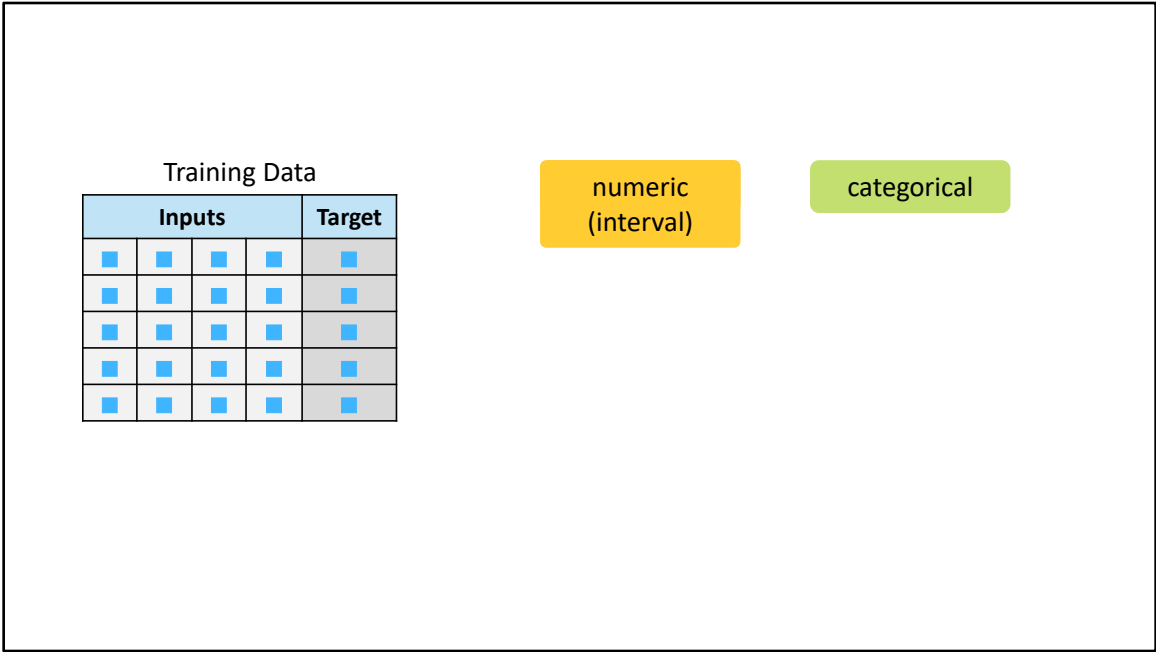


The variables include *inputs* and a *target*. In the data set used for the churn business challenge, some of the inputs are number of calls to technical support, account tenure, age of customer's device, and device manufacturer. The target indicates whether customers churned.

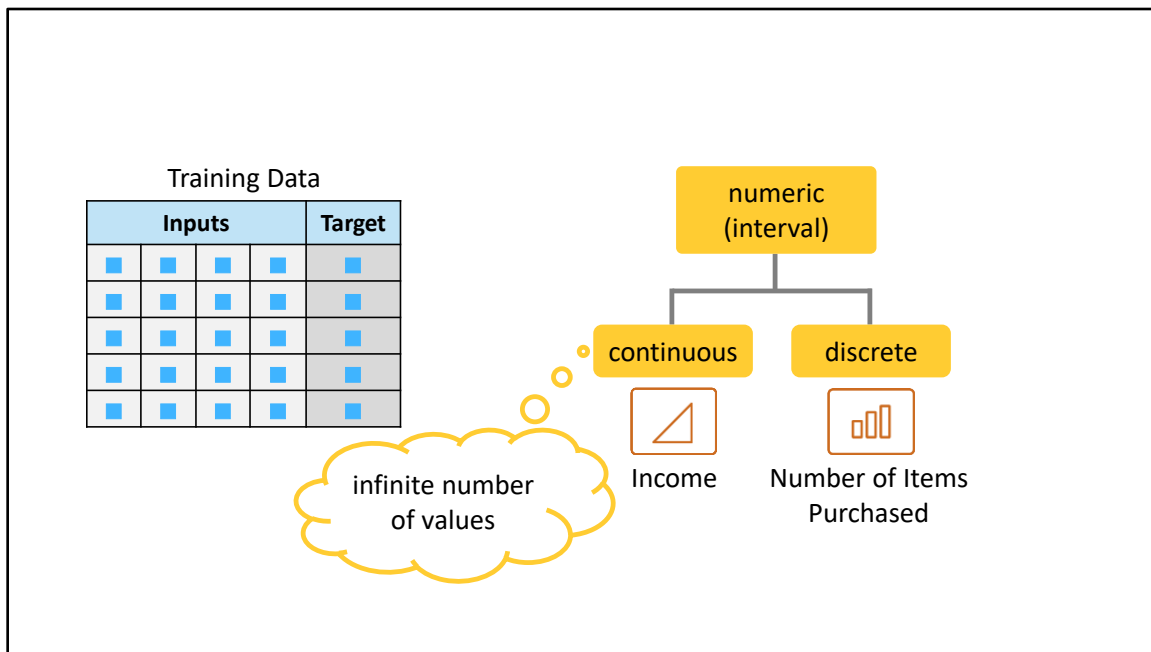
Inputs are also known as *predictors*, *features*, *explanatory variables*, or *independent variables*.

The *target* is also known as a *response*, *outcome*, or *dependent variable*.

For a given case, the inputs reflect your knowledge before measuring the target.

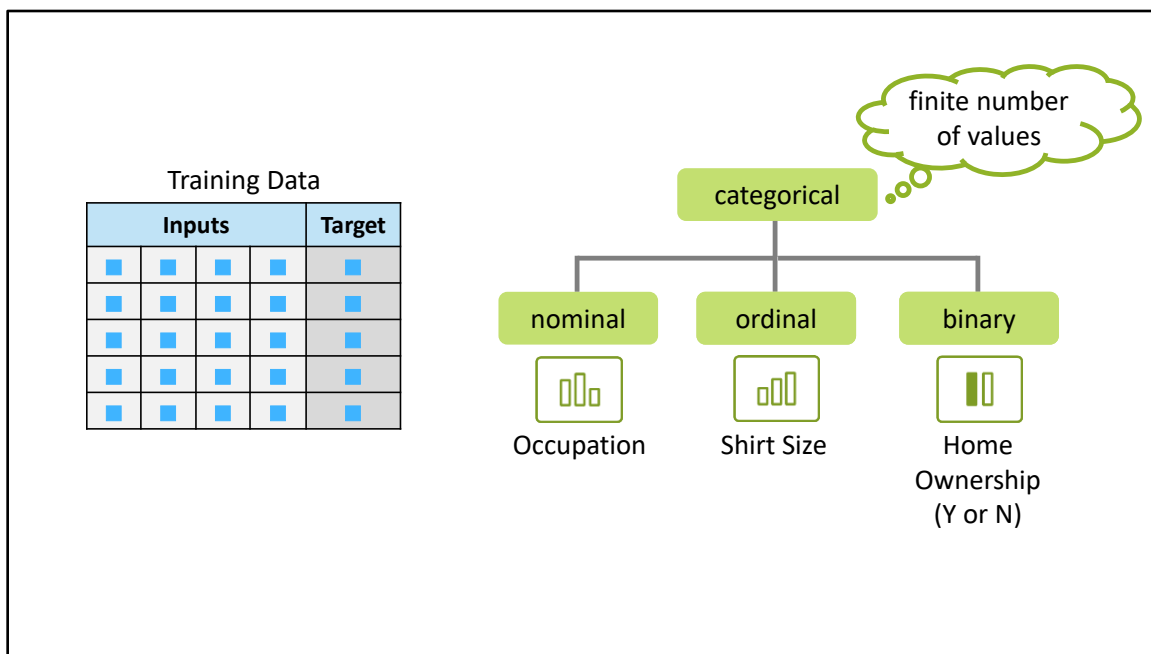


A variable can be numeric or categorical. Numeric variables are also known as interval variables.



Numeric variables can further be classified as continuous or discrete. Continuous variables can take on any one of an infinite number of values within a given range. Income is an example of a continuous variable. Of course, for practical reasons, income is typically rounded to no more than two decimal places.

Discrete variables are counts. The number of items purchased is an example of a discrete variable.

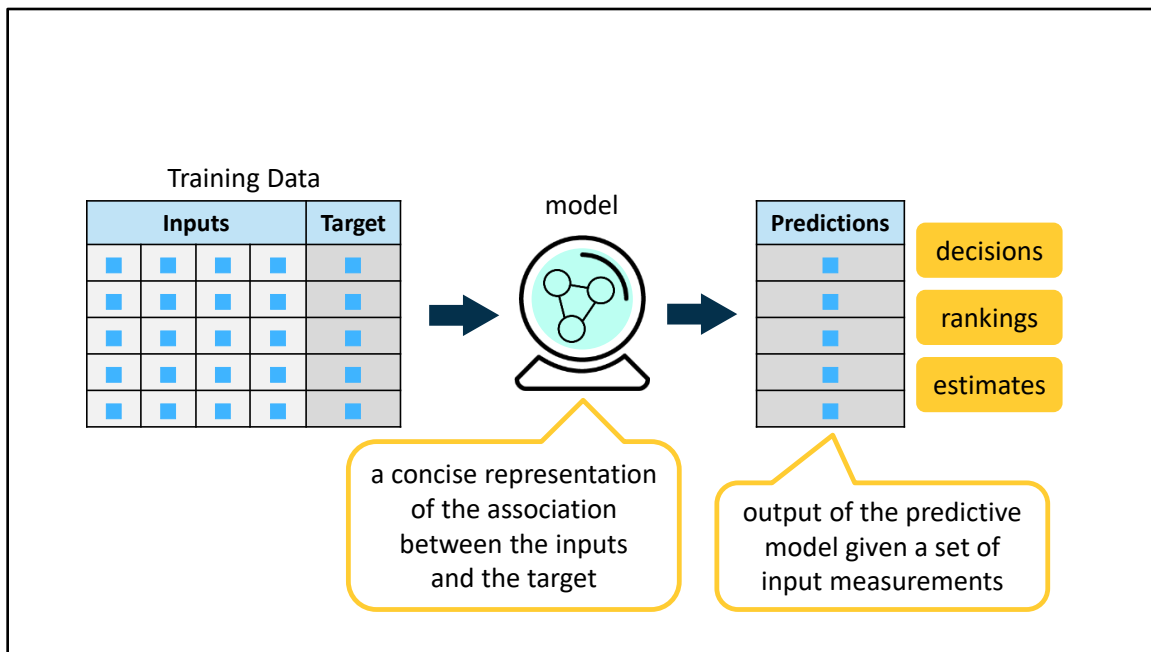


Categorical variables take on qualitative values, each value representing a group or category. The number of possible values of a categorical variable is typically finite.

Categorical variables are often nominal, which means that the categories have no particular order. Occupation is an example of a nominal variable.

Ordinal variables have categories with an inherent order (for example, Shirt Size).

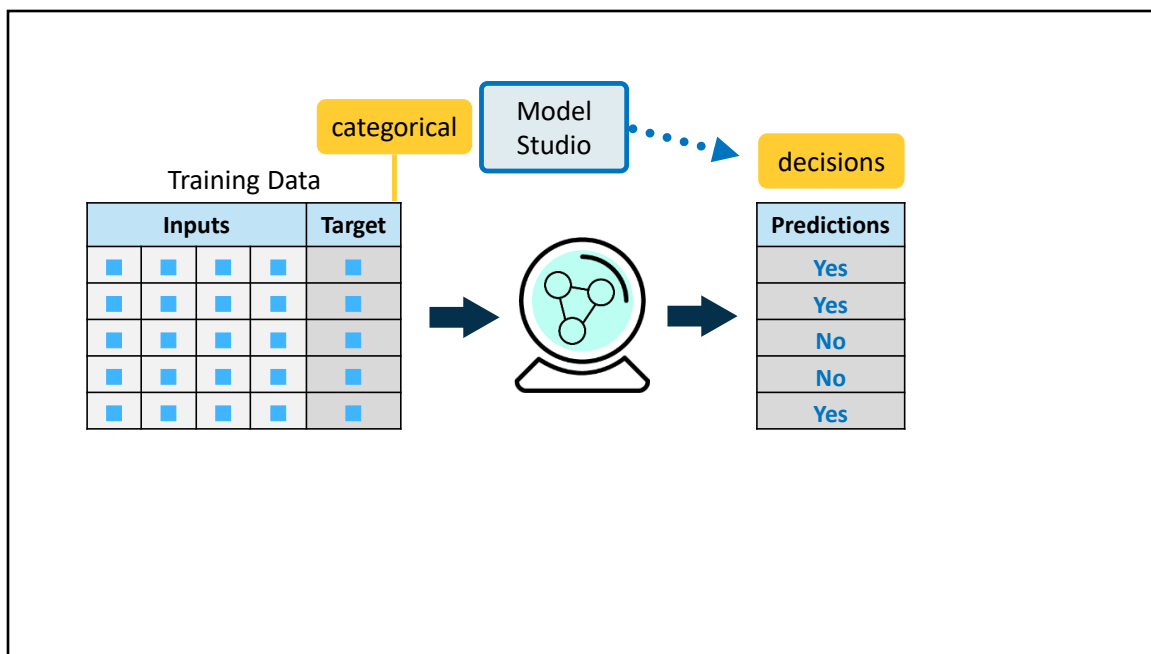
Binary variables have two categories, such as a positive or negative response concerning home ownership.



The purpose of the training data is to construct a predictive model (that is, a rule) that relates the inputs to the target. The *predictive model* is a concise representation of the association between the inputs and the target.

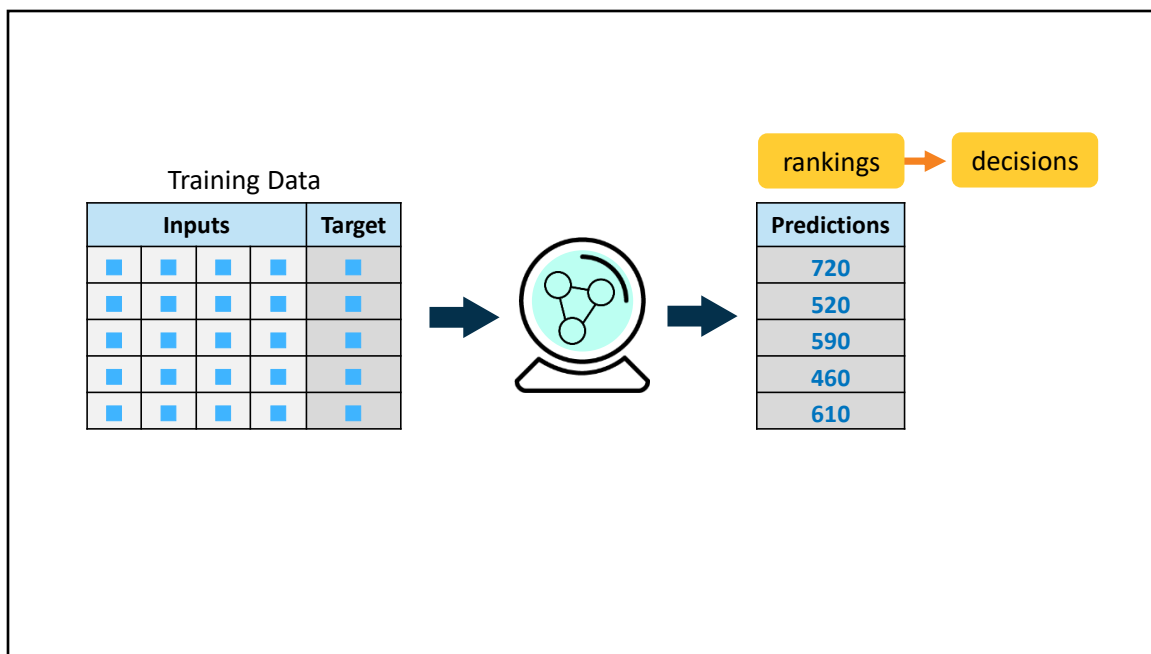
The outputs of the predictive model are known as *predictions*. Predictions represent the model's best guess about the target given a set of input measurements. The predictive model makes predictions based on what it learns from the source data.

A predictive model can generate one of three types of predictions: decisions, rankings, and estimates.



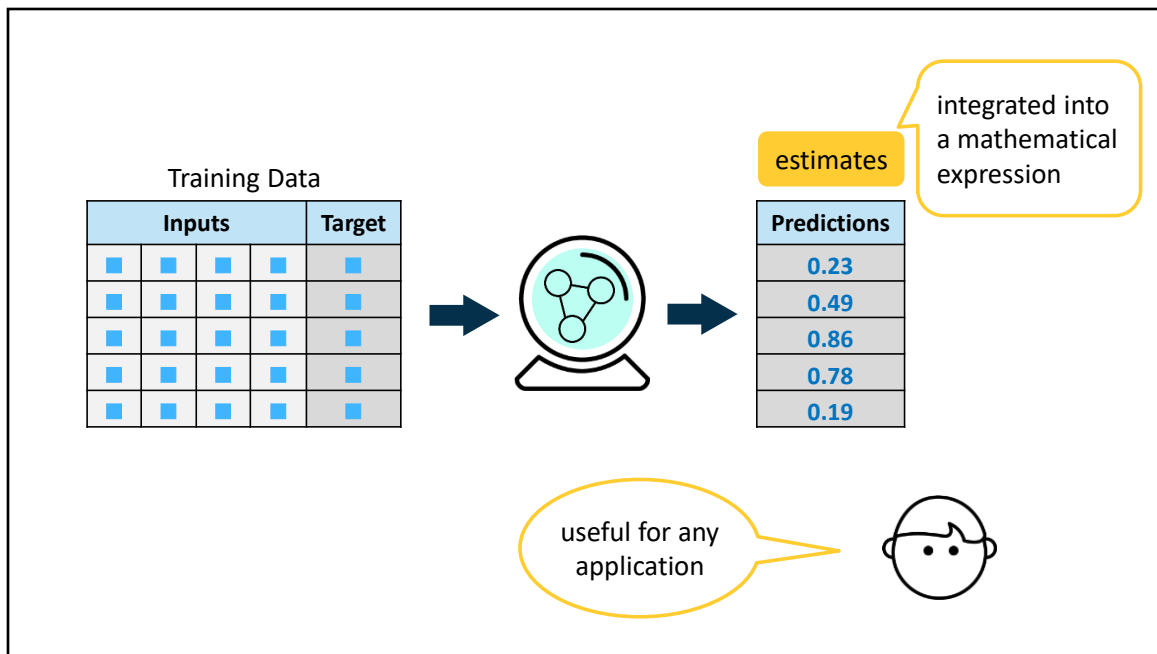
Decision predictions are the simplest type of prediction. Decisions are usually associated with some type of action (for example, classifying a case as a churn or a no-churn). For this reason, decisions are also known as *classifications*. Decision predictions are used, for example, in handwriting recognition, fraud detection, and direct mail solicitation.

Decision predictions usually relate to a categorical target variable. By default, Model Studio assumes decision predictions when the target variable is categorical (binary, nominal, or ordinal).



Ranking predictions order the cases based on the relationships between the input variables and the target variable. The most common example of a ranking prediction is a credit score. Using the training data, the model attempts to rank the high-value cases higher than the low-value cases. For example, the model might produce a list of customers ranked from highest priority to lowest priority. The actual rank values are irrelevant; only the relative order is important.

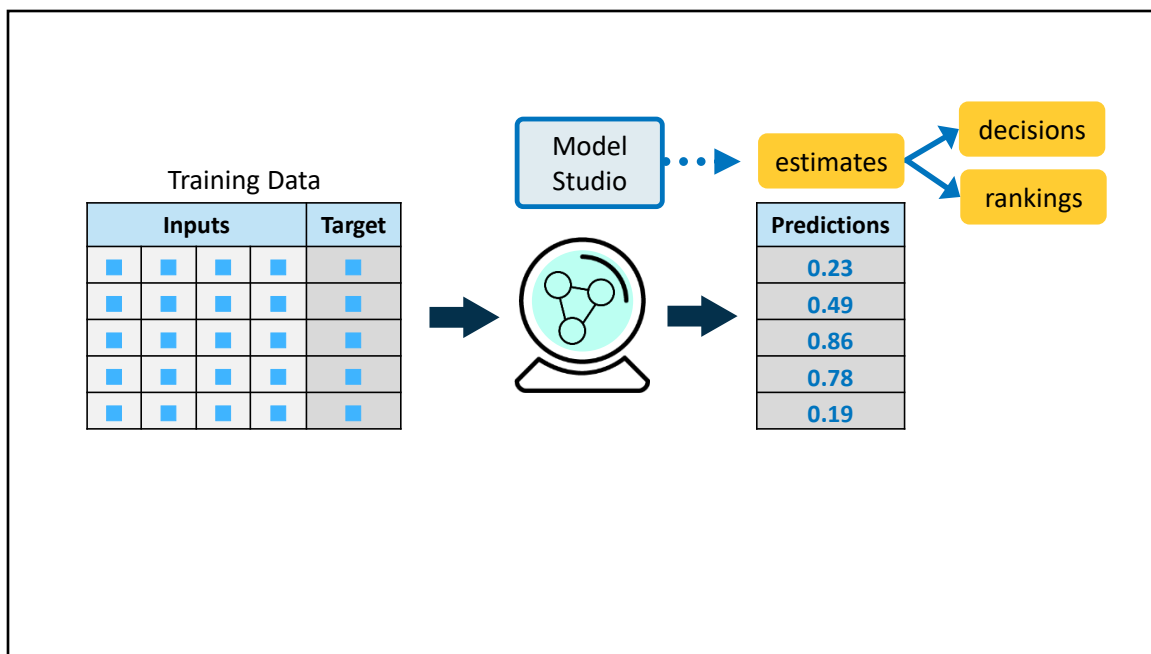
Note that ranking predictions can be transformed into decision predictions if cutoffs are defined for the ranks. For example, it is possible to make a binary decision by classifying all ranks above a cutoff as event cases and all ranks below the cutoff as nonevent cases.



Estimate predictions approximate the expected value of the target, given the input values. For cases with numeric targets, this estimate can be thought of as the average value of the target for all cases that have the observed input measurements. When the target is categorical, the estimate might equal the probability of a target outcome.

Prediction estimates are most commonly used when their values are integrated into a mathematical expression. An example is two-stage modeling, where the probability of an event is combined with an estimate of profit or loss to form an estimate of unconditional expected profit or loss.

Prediction estimates are also useful when you are not certain about the ultimate application of the model.



Estimate predictions are versatile because they can be transformed into both decisions and ranking predictions. Thus, if you are not sure which type of prediction to use, it is best to go with estimates.

In Model Studio, most types of models can be configured to produce estimate predictions.

Data Preparation and Preprocessing



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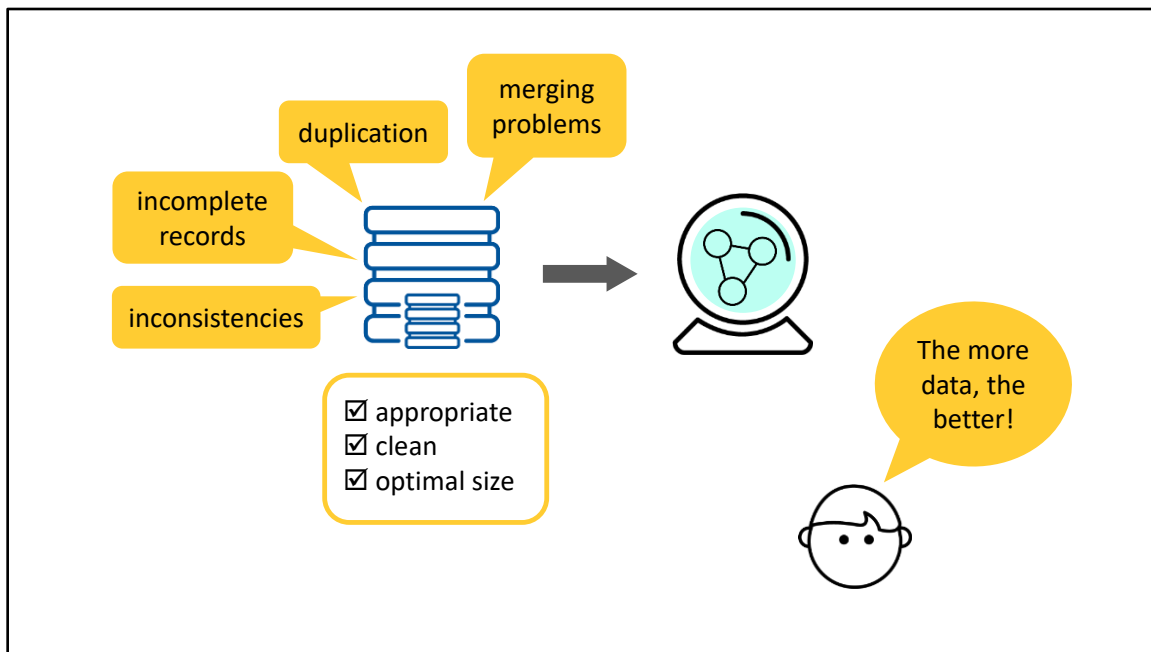
Data Preparation and Preprocessing



You start your analytics project at the Data phase of the analytics life cycle. This phase is also known as data preparation or data preprocessing.

Data preparation is the process of converting raw data into a wide table by parsing and joining data. At the end of data preparation, data are aggregated and summarized to the right granularity. Extraneous columns have been dropped, and invalid records have been filtered out. However, you are yet not ready to apply machine learning models.

The data now need to be preprocessed to achieve certain modeling requirements. Some common data preprocessing tasks are case filtering, value replacement, imputing missing values, feature scaling, truncating values, and one-hot-encoding categorical features. In this course, we focus on data preprocessing tasks.



Analysts generally agree that the more data you have, the better your models will be.

However, data preprocessing involves more than simply collecting a large amount of data. For one thing, data are usually dirty and noisy. It's typical to encounter inconsistencies, incomplete records, duplication, and merging problems.

Good data are the foundation for good models, so it is important to make sure that your data are appropriate, clean, and reduced to the optimal size for analysis. In fact, it is not unusual for analysts to spend the majority of their project time on data preprocessing.



Essential Data Tasks

- Divide the data.
- Address rare events.
- Manage missing values.
- Replace incorrect values.
- Add unstructured data.
- Extract features.
- Handle extreme or unusual values.
- Select useful inputs.

To prepare your data, you perform several essential data tasks. In this lesson, you learn about the first three tasks in the list: dividing the data, addressing rare events, and managing missing values. You perform these tasks while you create your Model Studio project and your first few pipelines. In the next lesson you learn about the remaining essential data tasks: replacing incorrect values, adding unstructured data, extracting features, handling extreme or unusual values, and selecting useful inputs.

Although you perform these tasks in the listed sequence in this course, keep in mind that you can vary the order in your own projects, as needed.

SAS Viya Tools for Data Preparation

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Dividing the Data



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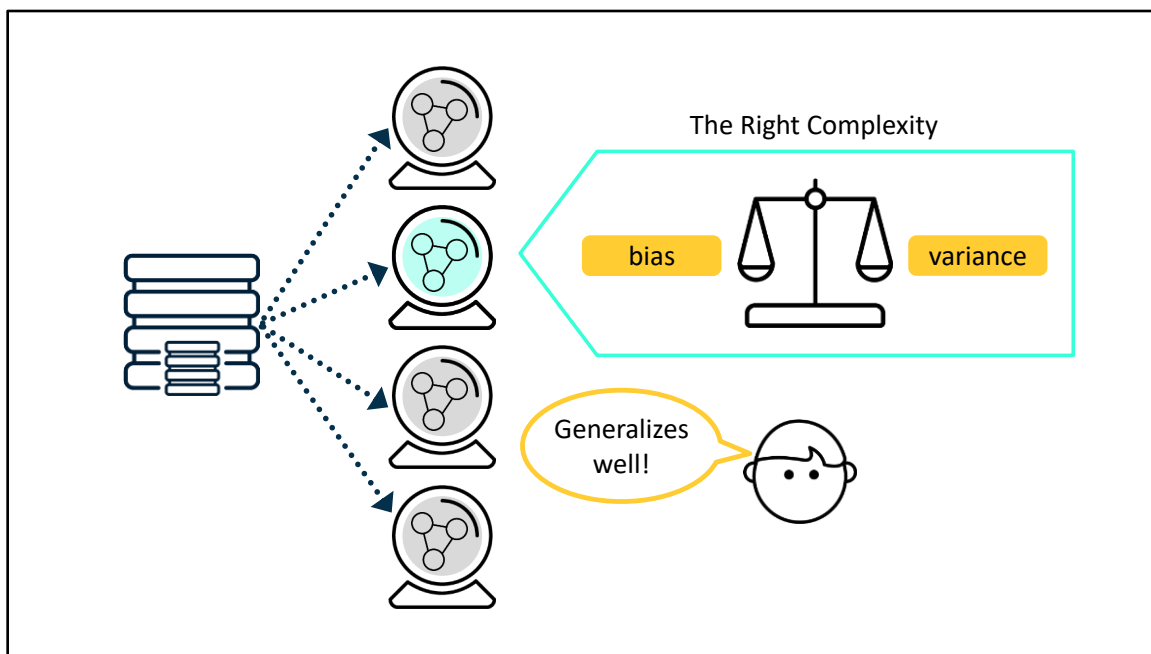
Dividing the Data



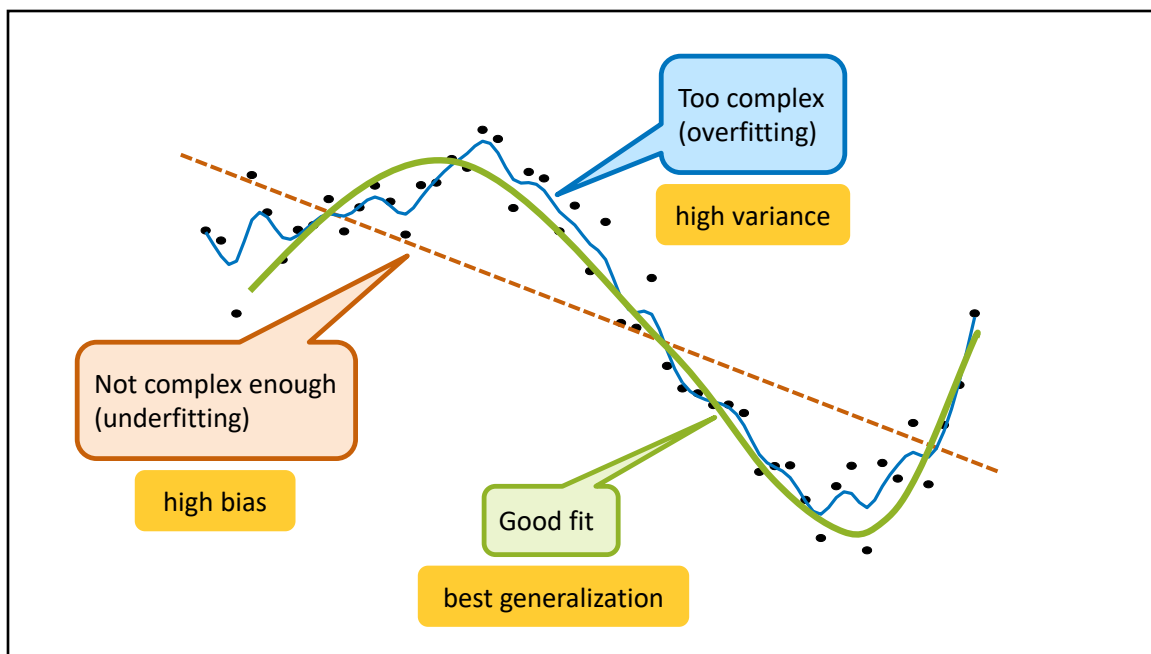
Essential Data Tasks

- Divide the data.
- Address rare events.
- Manage missing values.
- Replace incorrect values.
- Add unstructured data.
- Extract features.
- Handle extreme or unusual values.
- Select useful inputs.

We're ready for the first essential data task: dividing the data. To start, let's see why it's a good idea to divide the data.



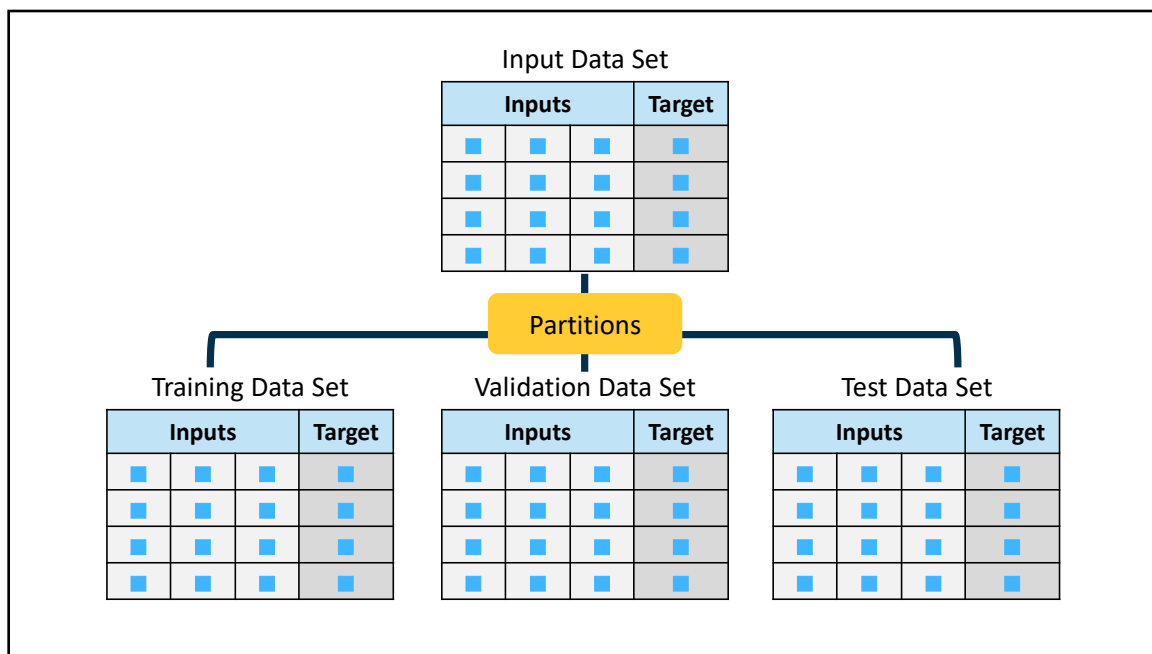
Fitting a model to data requires searching through the space of possible models, building a series of models of increasing complexity, and selecting a model that generalizes well. You want to choose the right complexity, which involves a trade-off between bias and variance.



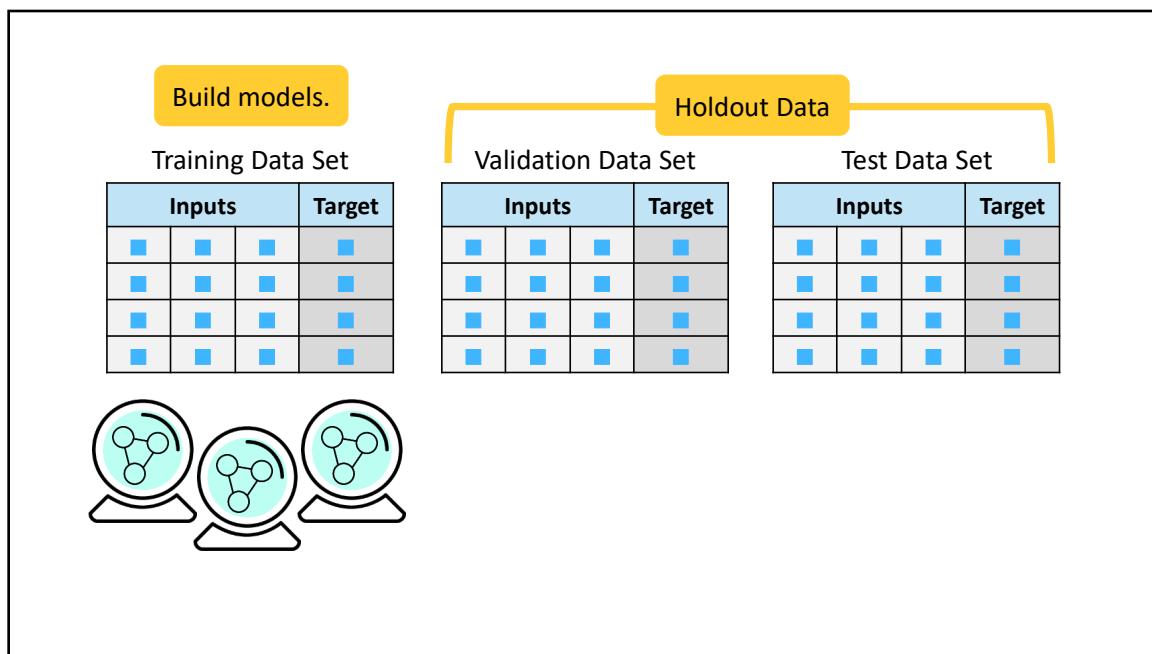
Suppose you have some data points, as shown here. An insufficiently complex model might not be flexible enough, which can lead to underfitting (that is, systematically missing the signal). In this situation, the model has high bias.

A naïve modeler might assume that the most complex model should always outperform the others, but this is not the case. An overly complex model might be too flexible, which can lead to overfitting (that is, accommodating nuances of the random noise in the sample). In this situation, the model has high variance.

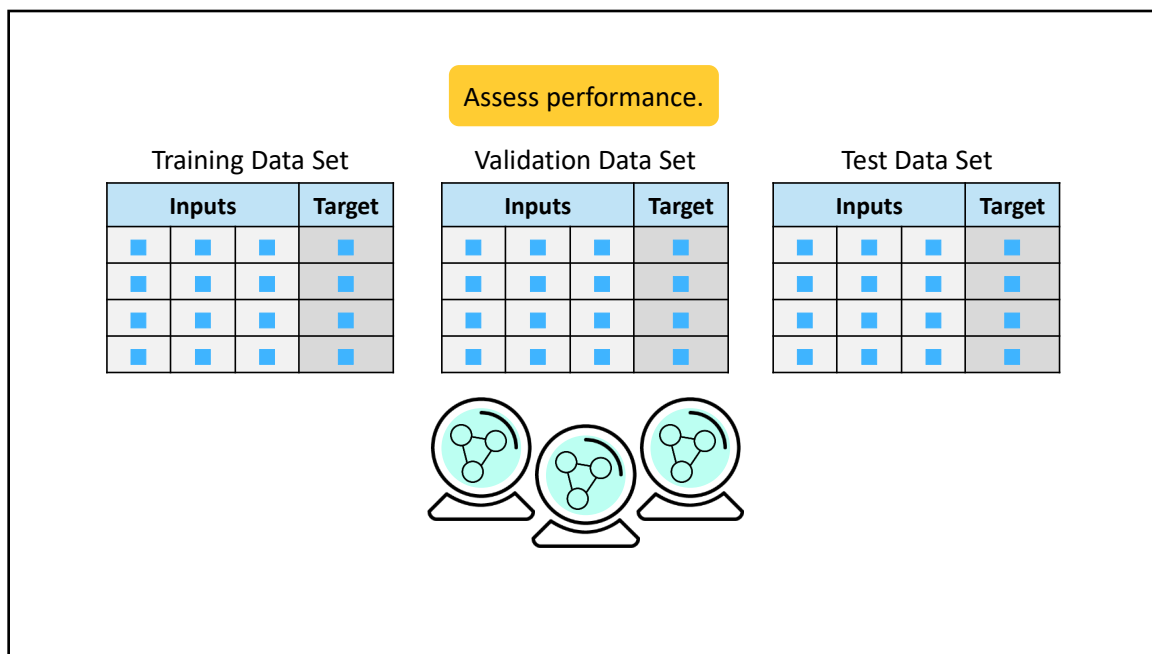
The goal of model selection is to choose a model that neither overfits nor underfits the data. A model with the right amount of flexibility gives the best generalization.



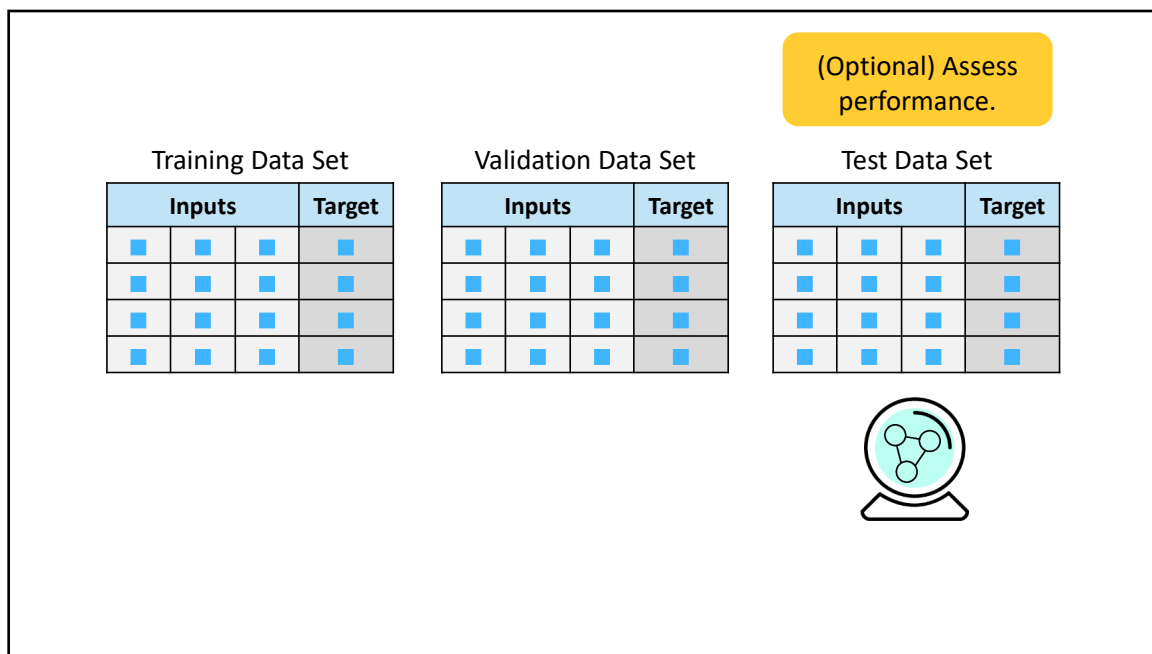
To allow for honest assessment of model performance, the data are typically divided into two or three non-overlapping sets, which are called partitions.



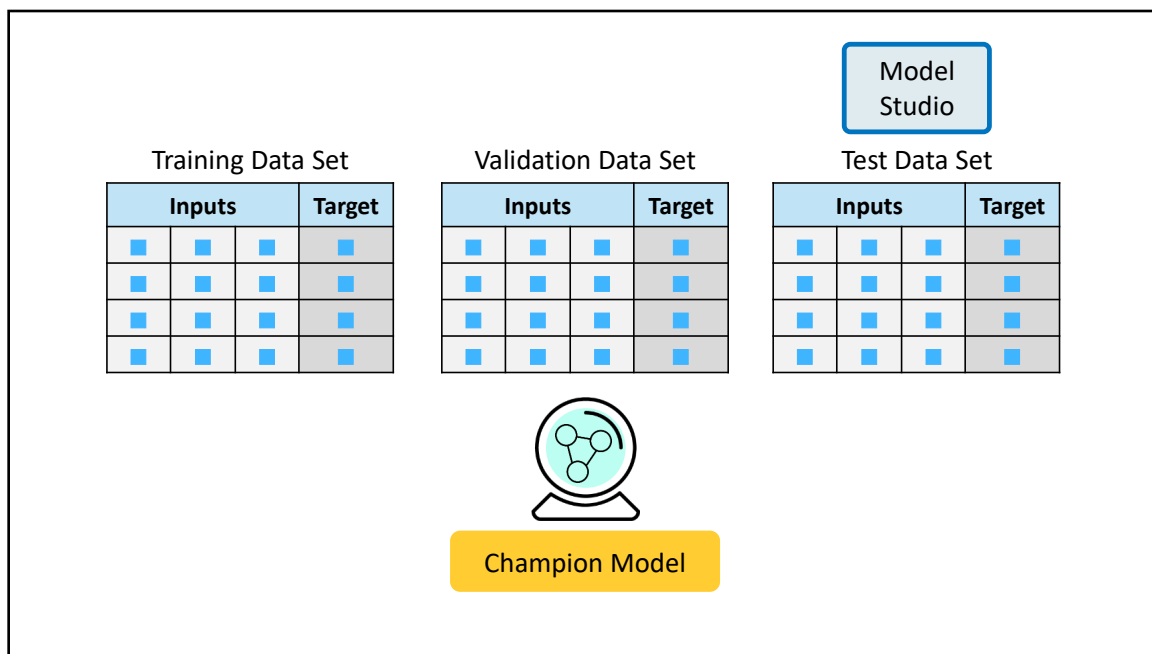
You use the first partition, the *training set*, to build models. Remaining partitions of the data can be referred to as holdout data.



After model training, you can assess the performance of each model on the second partition of the data, the *validation set*. The validation data set is used to optimize the complexity of the model and find the sweet spot between bias and variance. Based on the validation data, you tune the models that were built on the training data and determine whether additional training is required.



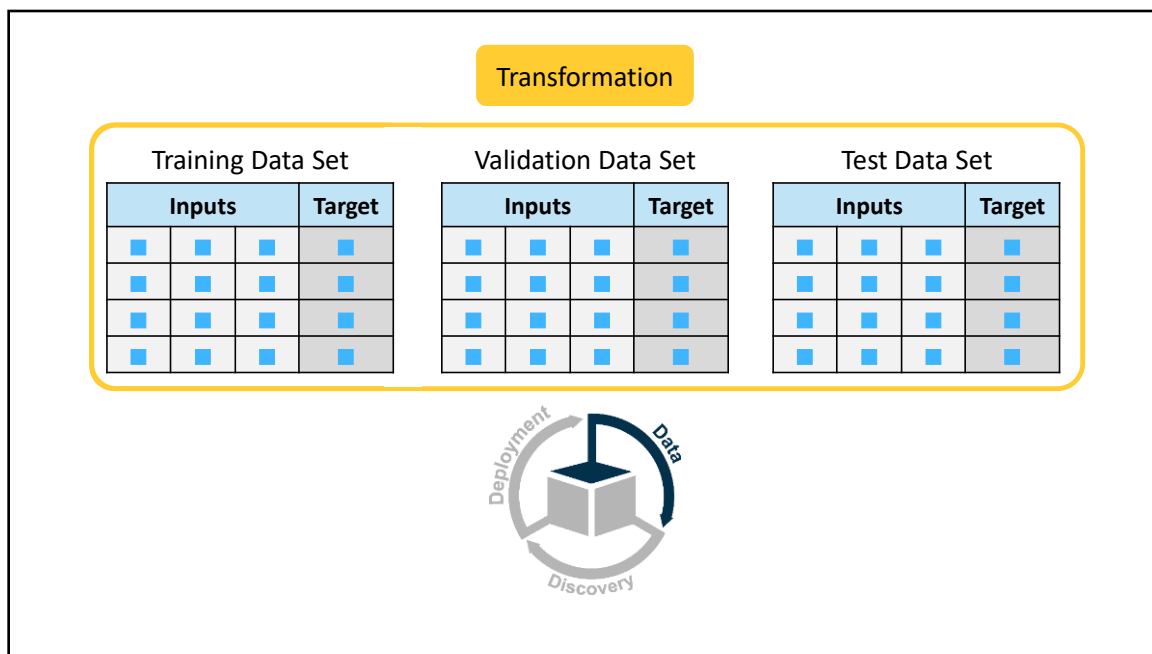
The *test data set* is an optional partition for the model building process, but some industries might require it as a source of unbiased model performance. The test data set gives the honest, unbiased estimates of the model's performance. The test data set provides one final measure of how the model performs on new data, before the model is put into production.



By default, Model Studio uses the test data to select the champion model, if a test data set exists.

Otherwise, Model Studio uses the validation data to select the champion model.

If a validation data set does not exist, Model Studio uses the training data to select a champion model.



Many of the essential data tasks involve transforming the data. It is critical that all transformations that are used to prepare the training data are also applied to any holdout data.

Cross Validation for Small Data Sets

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Addressing Rare Events Using Event-Based Sampling



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Addressing Rare Events Using Event-Based Sampling



Essential Data Tasks

- ✓ Divide the data.
- Address rare events.
- Manage missing values.
- Replace incorrect values.
- Add unstructured data.
- Extract features.
- Handle extreme or unusual values.
- Select useful inputs.

It is important to determine whether your target can be characterized as a rare event relative to the total number of cases.

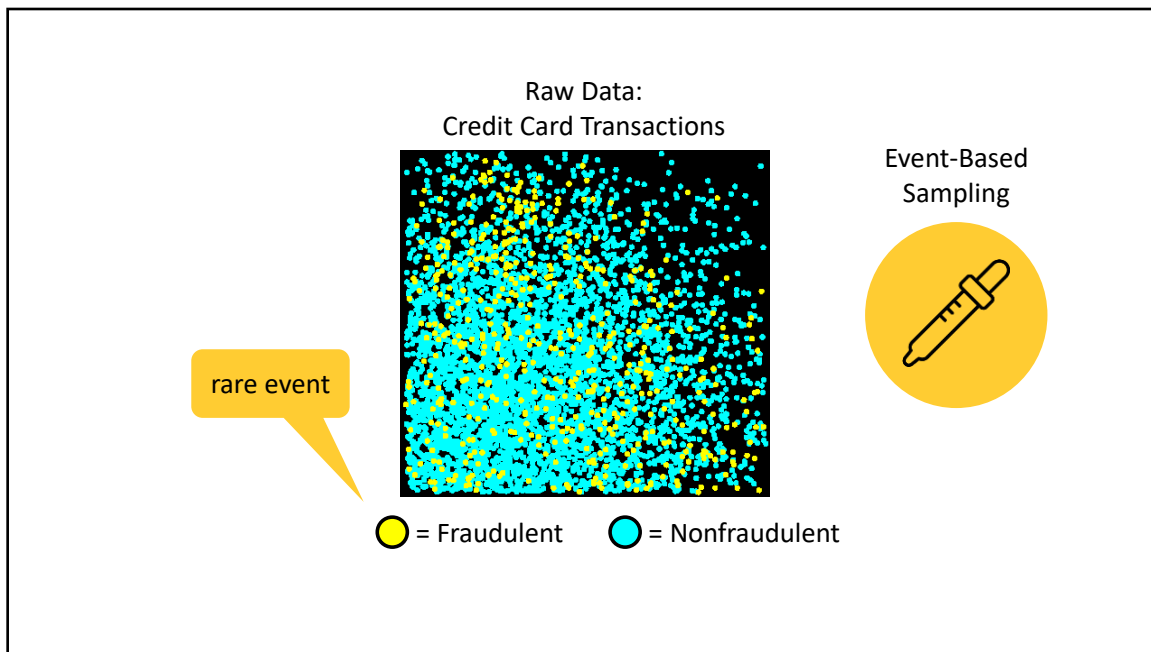


Training Data

Inputs				Target
■	■	■	■	■
■	■	■	■	■
■	■	■	■	■
■	■	■	■	■
■	■	■	■	■

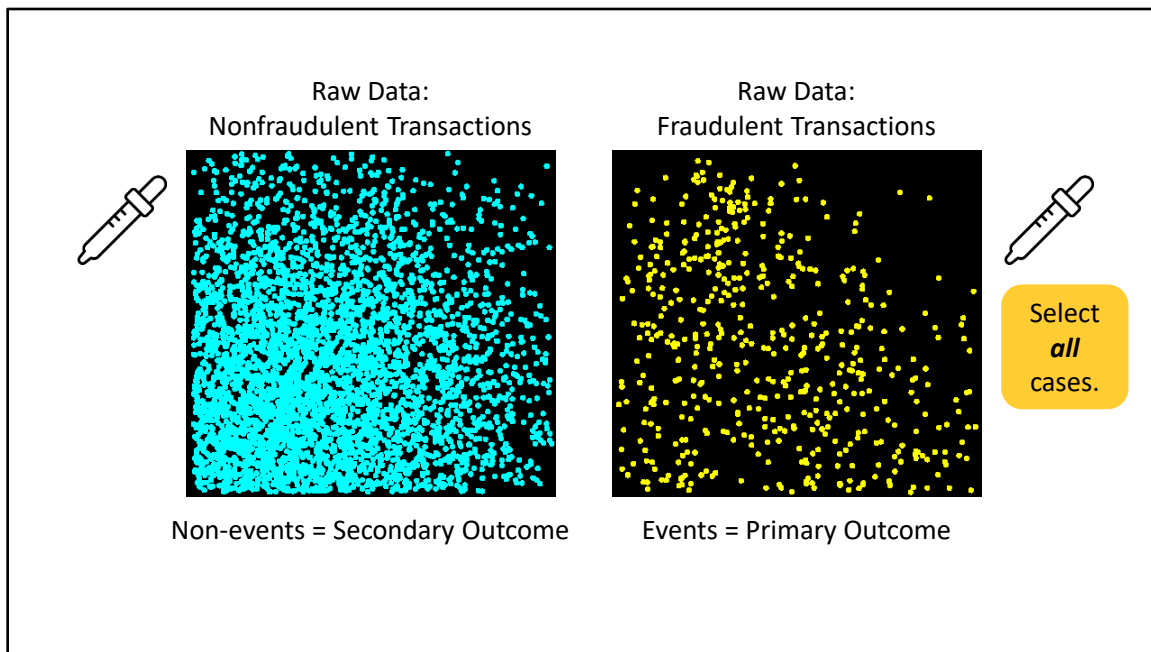
Include a representative number of fraudulent cases.

For example, in the detection of fraudulent activity, it is necessary to ensure that the training data include a representative number of fraudulent cases so as to capture the event sufficiently.

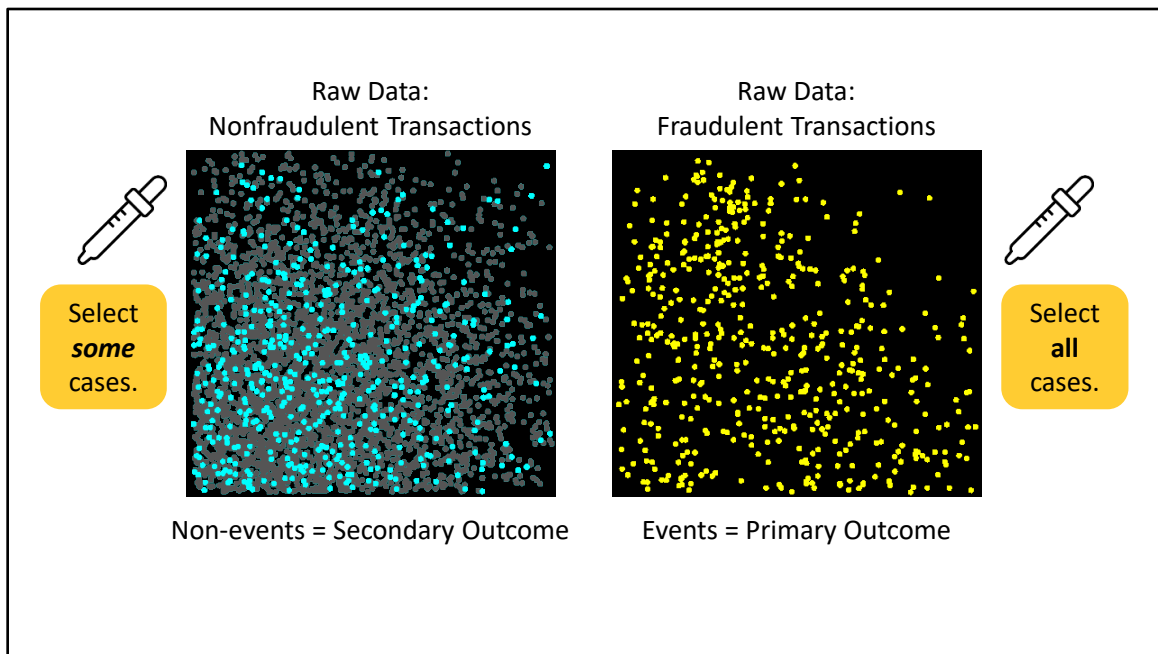


One out of every 1,000 credit card transactions is fraudulent, so fraudulent transactions are a rare event. Fitting a model to such data without accounting for the extreme imbalance in the occurrence of the event does not generate useful results.

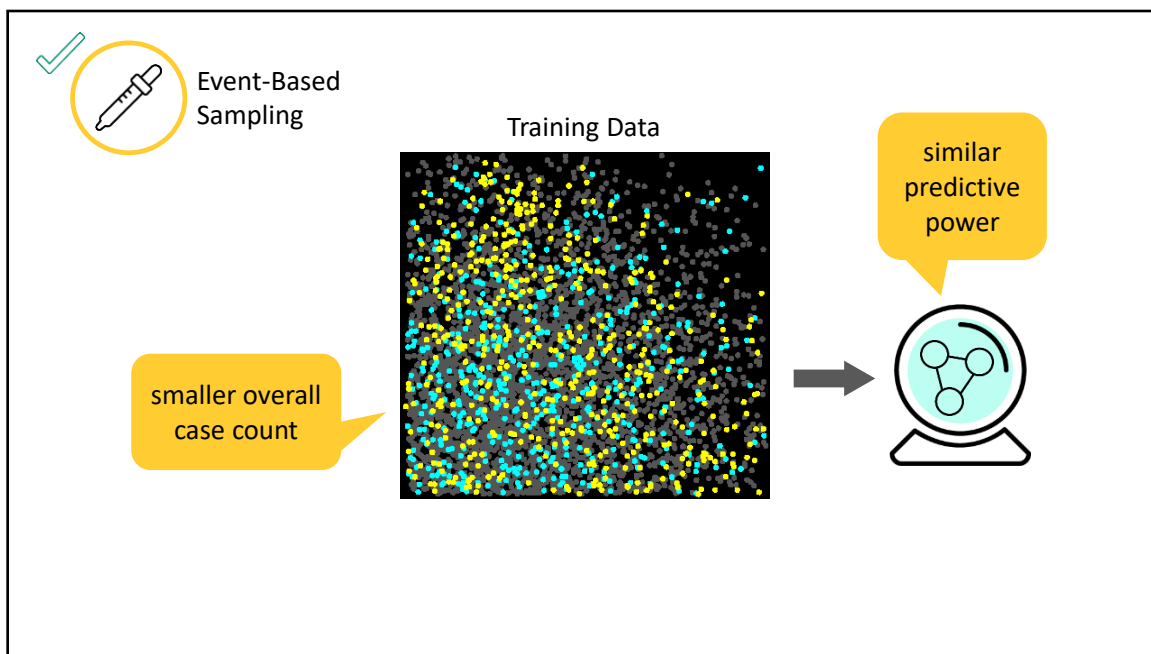
Special sampling methods are commonly used to modify an imbalanced data set (that is, a data set with a rare event) and provide a more balanced distribution. To address a rare target, a common predictive modeling practice is to build models from a sample with a primary outcome proportion that is different from the true population proportion. This technique, called event-based sampling, is typically used when the ratio of primary to secondary outcome cases is small.



In event-based sampling, samples are drawn separately based on the target events (that is, the primary outcome cases) and the non-events (that is, the secondary outcome cases). If the event is rare, usually all events are selected.

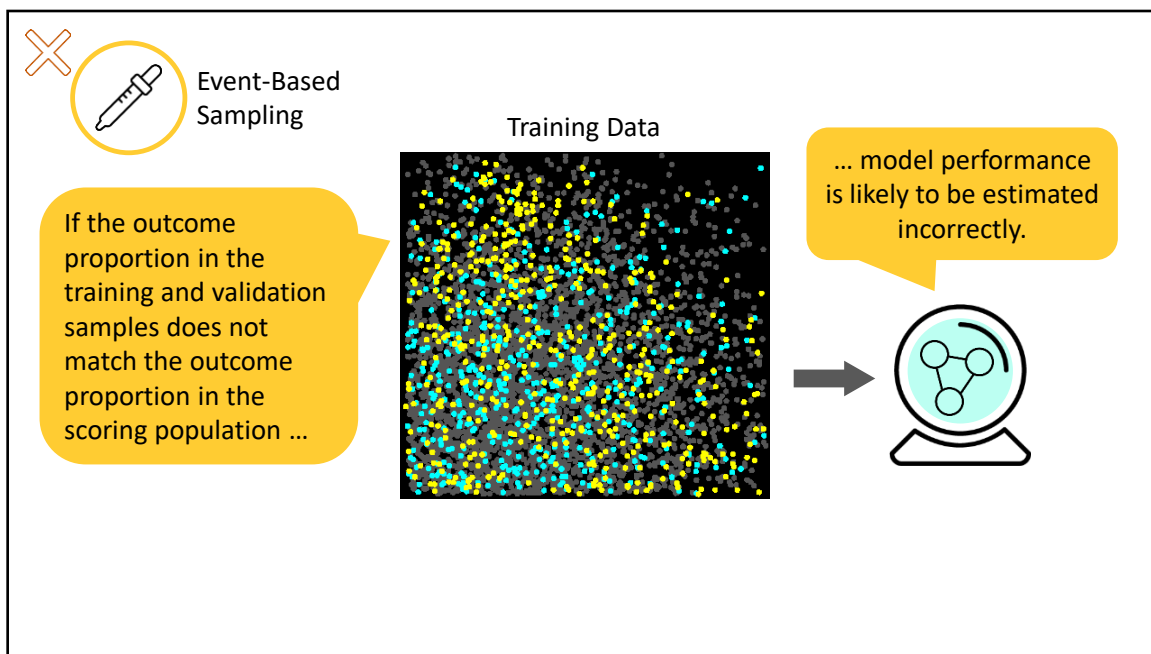


Then each event outcome is matched by one or more non-event outcomes. In other words, only some of the non-event cases are selected.



The advantage of event-based sampling is that you can obtain (on the average) a model of similar predictive power with a smaller overall case count. This is in concordance with the idea that the amount of information in a data set with a categorical outcome is determined not by the total number of cases in the data set itself, but instead by the number of cases in the rarest outcome category. For binary target data sets, this is usually the event outcome.

This advantage might seem of minimal importance in the age of extremely fast computers. However, model fitting occurs only after the completion of a long, tedious, and error-prone data preparation process. Smaller sample sizes for data preparation are usually welcome.



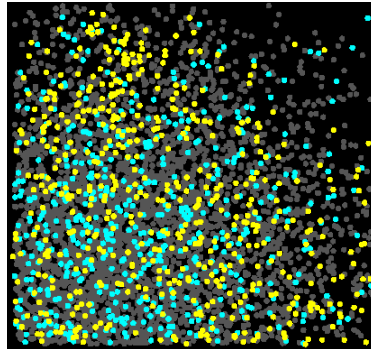
Although it reduces analysis time, event-based sampling also introduces some analysis complications.

Most model fit statistics (especially those related to prediction decisions) and most of the assessment plots are closely tied to the outcome proportions in the training samples. If the outcome proportion in the training and validation samples does not match the outcome proportion in the scoring population, model performance is likely to be estimated incorrectly. If the outcome proportions in the training sample and scoring population do not match, model prediction estimates are biased.



Event-Based
Sampling

Training Data



adjusts scoring
results for bias

Model
Studio



Fortunately, Model Studio automatically adjusts assessment measures, assessment graphs, and prediction estimates for bias if you enable event-based sampling.

Demo: Modifying the Data Partition

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Selecting Variables on the Data Tab

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Question: Data Partitions



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Question: Data Partitions

When there is a training data set, a validation data set, and a test data set, which data partition is used by default to select the champion model?

- a. training data set
- b. validation data set
- c. test data set

Answer: Data Partitions

When there is a training data set, a validation data set, and a test data set, which data partition is used by default to select the champion model?

- a. training data set
- b. validation data set
- ☒ c. test data set

CA: c

FB: If the test data set exists, it is used by default to select the champion model. If only training and validation data sets exist, then the validation data set is used to select the champion model. If only a training data set exists, it is used to select the champion.

Global Metadata

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Managing Missing Values



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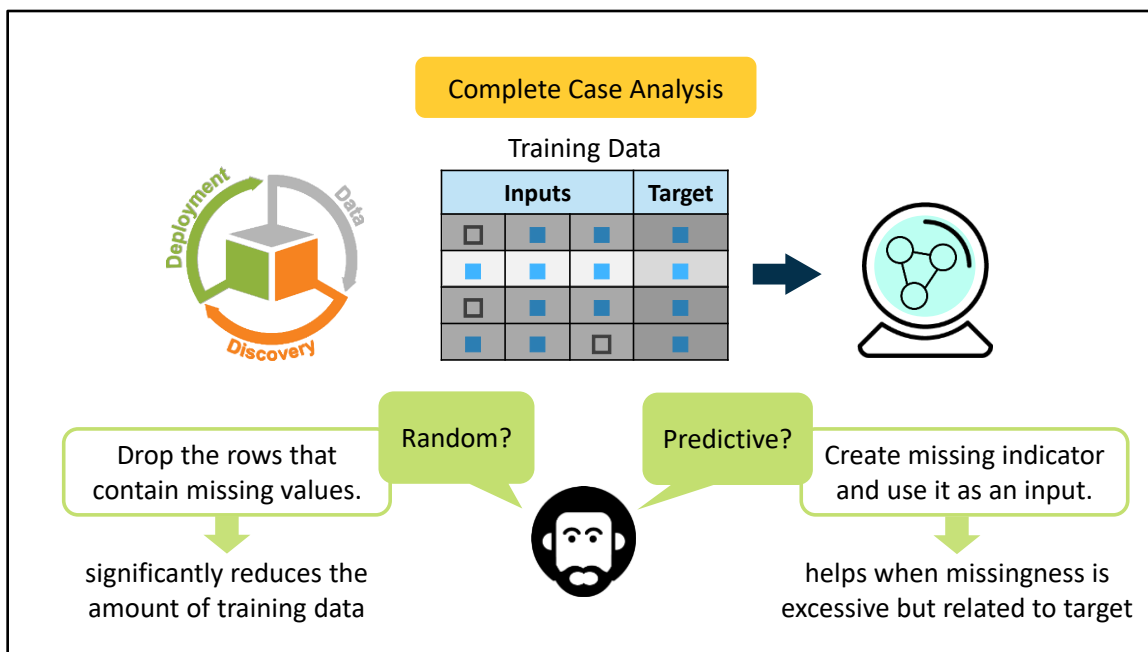
Managing Missing Values



Essential Data Tasks

- ✓ Divide the data.
- ✓ Address rare events.
- **Manage missing values.**
- Replace incorrect values.
- Add unstructured data.
- Extract features.
- Handle extreme or unusual values.
- Select useful inputs.

The data that you use for predictive modeling are likely to have missing values. It's best to handle the problem of missing values early in data preprocessing.



Missing values might occur in the target variable or the input variables. It is especially problematic for your analysis when the target has missing values. However, in this course, you learn about the more common scenario of missing values in the input variables.

Missing values can be a problem during both model building (in the Discovery phase) and model deployment. Let's see why.

To build a model, several machine learning algorithms in SAS Viya use only the complete cases (that is, the cases with no missing values). With this approach, which is called complete case analysis, the model might be based on only a small subset of the original data.

It is important to consider whether missing values are distributed randomly or whether missingness is somehow predictive of the target. If missing values appear at random in the input data, you can drop the rows that contain missing values without introducing bias into the model. However, complete case analysis can remove a tremendous amount of information from the training data and reduce the predictive accuracy of the model.

Cases often contain missing values for a reason. If the missingness itself is predictive, that is, if the missing value is in some way related to the target variable, then the variable might be better handled by creating the binary missing indicator and use it as an input to the model.



Scoring Data

Inputs			Target

It is very likely that your scoring data will also have the same type of missingness.

If your training data has missing values, it is very likely that your scoring data will also have missing values with the same type of missingness (that is, either random or predictive).



Scoring Data

Inputs			Target

$$\text{logit}(p) = -0.81 + 0.92 \cdot x_1 + 1.11 \cdot x_2$$

Predict: $(x_1, x_2) = (0.3, ?)$

Prediction formula cannot score cases with missing values.

In deployment, the model cannot score a case that has missing values.



Training Data

Inputs			Target
□	■	■	■
■	■	■	■
□	■	■	■
■	■	□	■



managing missing values when training a model



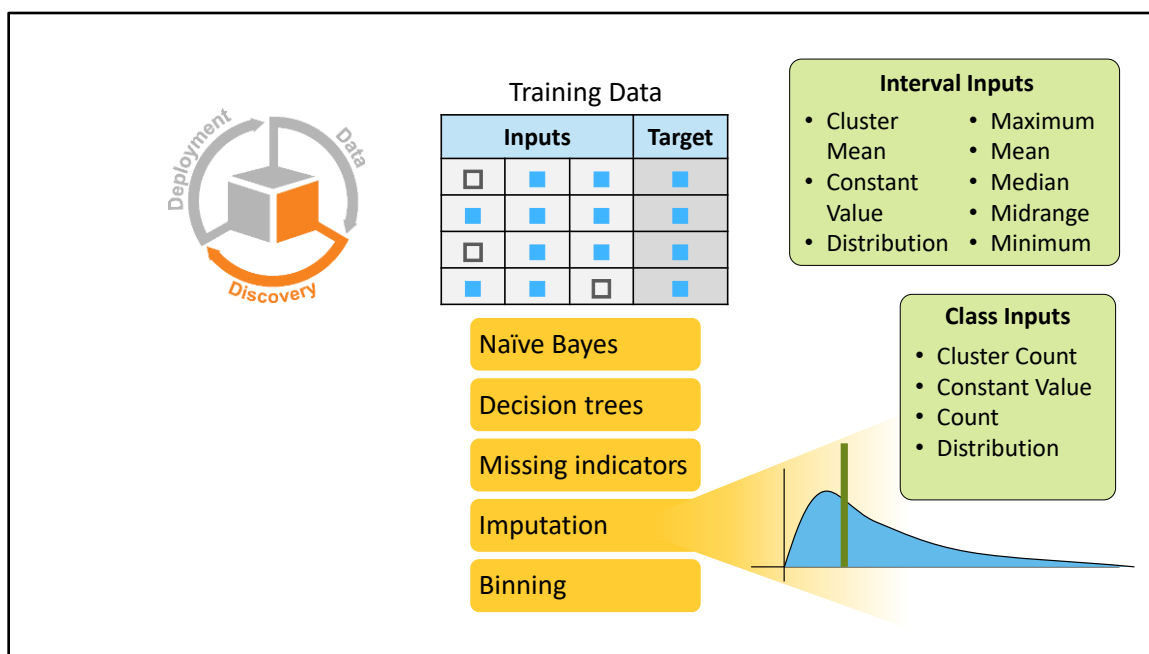
Scoring Data

Inputs			Target
□	■	■	■
■	■	■	■
□	■	■	■
■	■	□	■



handling missing values when scoring a new data

In this course, you learn how to manage missingness in training a machine learning model. You also learn guidelines for handling missing values when scoring data.



Techniques for managing missingness during model building include naïve Bayes, decision trees, missing indicators, imputation, and binning. Remember that missingness can be predictive. Thus, these techniques retain information that is associated with missing values, including the missing values themselves, which increases the predictive accuracy of the model.

In Model Studio, the naïve Bayes technique and decision tree models do not use complete case analysis; these modeling approaches can incorporate missingness.

Missing indicators are new inputs that indicate whether a value of a related variable is missing.

Imputation refers to replacing a missing value with information that is derived from nonmissing values in the training data. This is the approach that we'll use in this course.

Binning is helpful for handling interval input variables. In binning, the original numeric values are grouped into discrete categories called bins. The missing values are assigned to their own bin. The interval variable then becomes a categorical variable.

Next, you can learn additional details about these methods.

Managing Missing Values: Details

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Imputation Methods in Model Studio

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Question: Missing Values



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Question: Missing Values

What is it called when the model-building process ignores training data cases with missing values of inputs?

- a. event-based sampling
- b. complete case analysis
- c. honest assessment
- d. imputation

Answer: Missing Values

What is it called when the model-building process ignores training data cases with missing values of inputs?

- a. event-based sampling
- ☒ b. complete case analysis
- c. honest assessment
- d. imputation

CA: b

FB: Complete case analysis means that training data cases with missing values of inputs are ignored.

Demo: Building a Pipeline from a Basic Template

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Automated Pipeline Creation

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Pipeline Templates in Model Studio

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Question: Pipeline Templates



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Question: Pipeline Templates

Which default model is in the basic template for class target?

- a. decision tree
- b. linear regression
- c. logistic regression
- d. neural network

Answer: Pipeline Templates

Which default model is in the basic template for class target?

- a. decision tree
- b. linear regression
- ☒ c. logistic regression
- d. neural network

CA: c

FB: The Basic template for class target uses a logistic regression model by default.

Logistic Regression

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1.3 A Closer Look at SAS Viya



SAS Cloud Analytic Services

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SAS Viya: A Shift in Mindset

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Data Sources and CAS

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Question: CASLIBs



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Question: Caslibs

What does a caslib do?

- a. enables model content and score code to be integrated with applications outside of SAS
- b. provides access to files in a data source and to in-memory tables
- c. secures network connections
- d. compares pipelines

Answer: Caslibs

What does a caslib do?

- a. enables model content and score code to be integrated with applications outside of SAS
- ☒ b. provides access to files in a data source and to in-memory tables
- c. secures network connections
- d. compares pipelines

CA: b

FB: A CASLIB provides access to files in a data source, such as a database or a file system directory, and to in-memory tables.